

Technical drawing of Segment 2, showing a cross-section of the vessel. The drawing includes various callouts for components and dimensions.

Dimensions:

- UNTANKING HEIGHT = 446.500
- 118.110
- 40.564
- 35.882
- 33.275

Component Callouts:

- 03
- 23
- 49
- 32
- 53
- 42
- 55
- 54
- 22
- 14
- 34
- 33
- 30
- 36
- 61
- 30

Detail View:

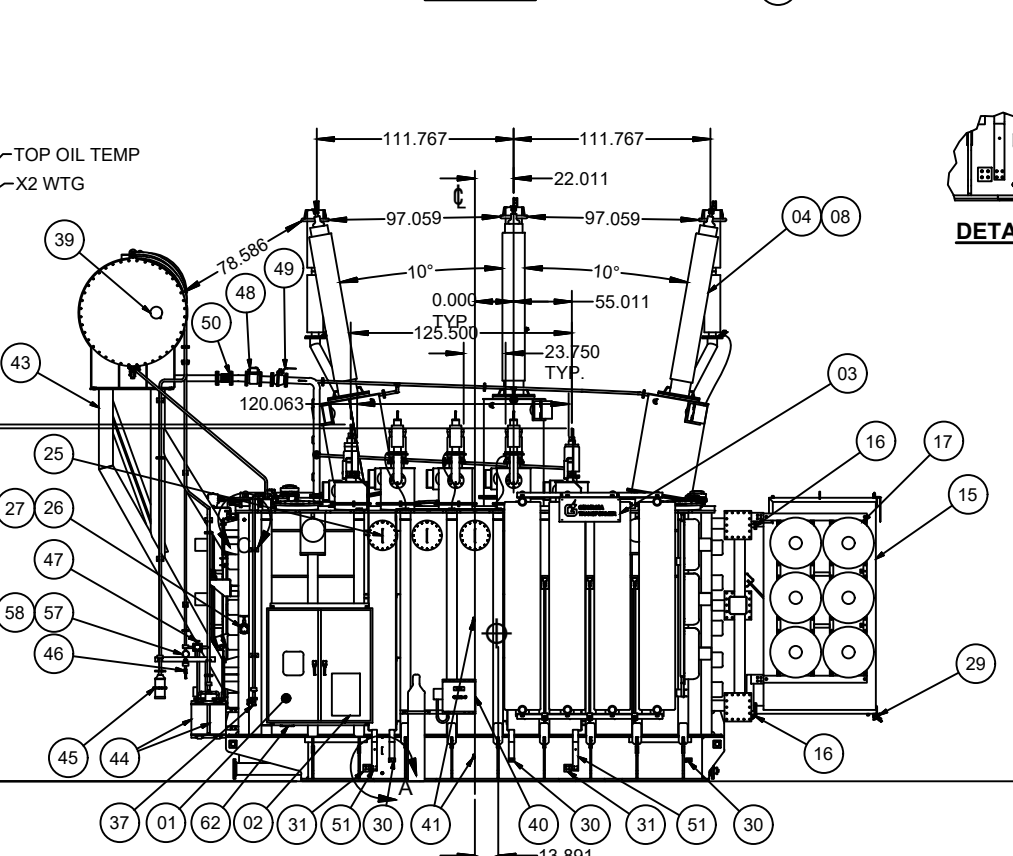
- H2 WTG
- 56

Height Specifications:

- HV NEUTRAL BUSHING HEIGHT = 202.637
- LV NEUTRAL BUSHING HEIGHT = 200.517

Segment Label:

SEGMENT 2



DETAIL C

SPARE THERMOWELL

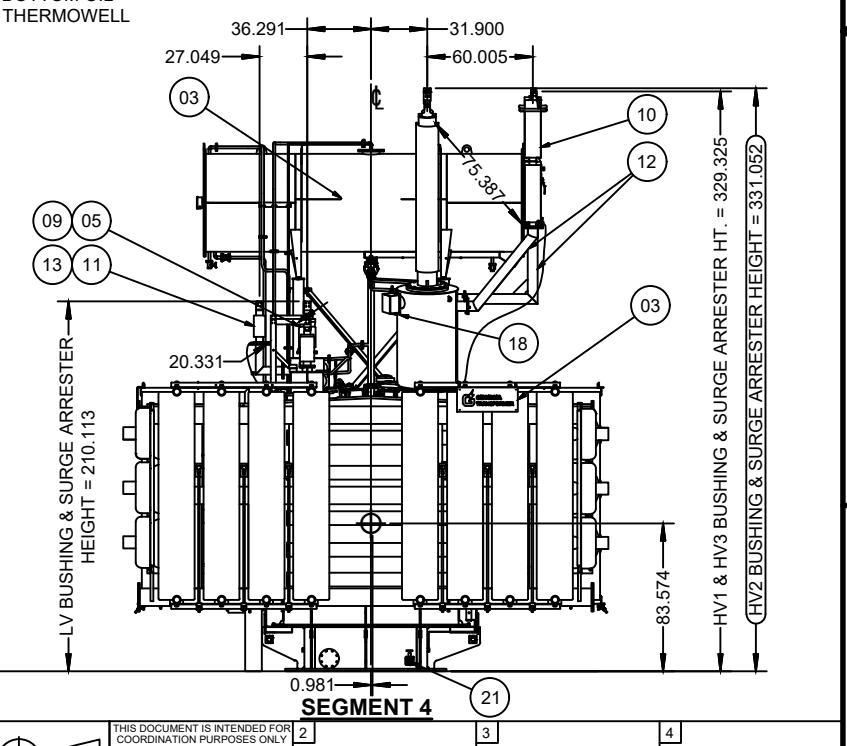
DETAIL D

LEGEND:

⋈ - CENTERLINE

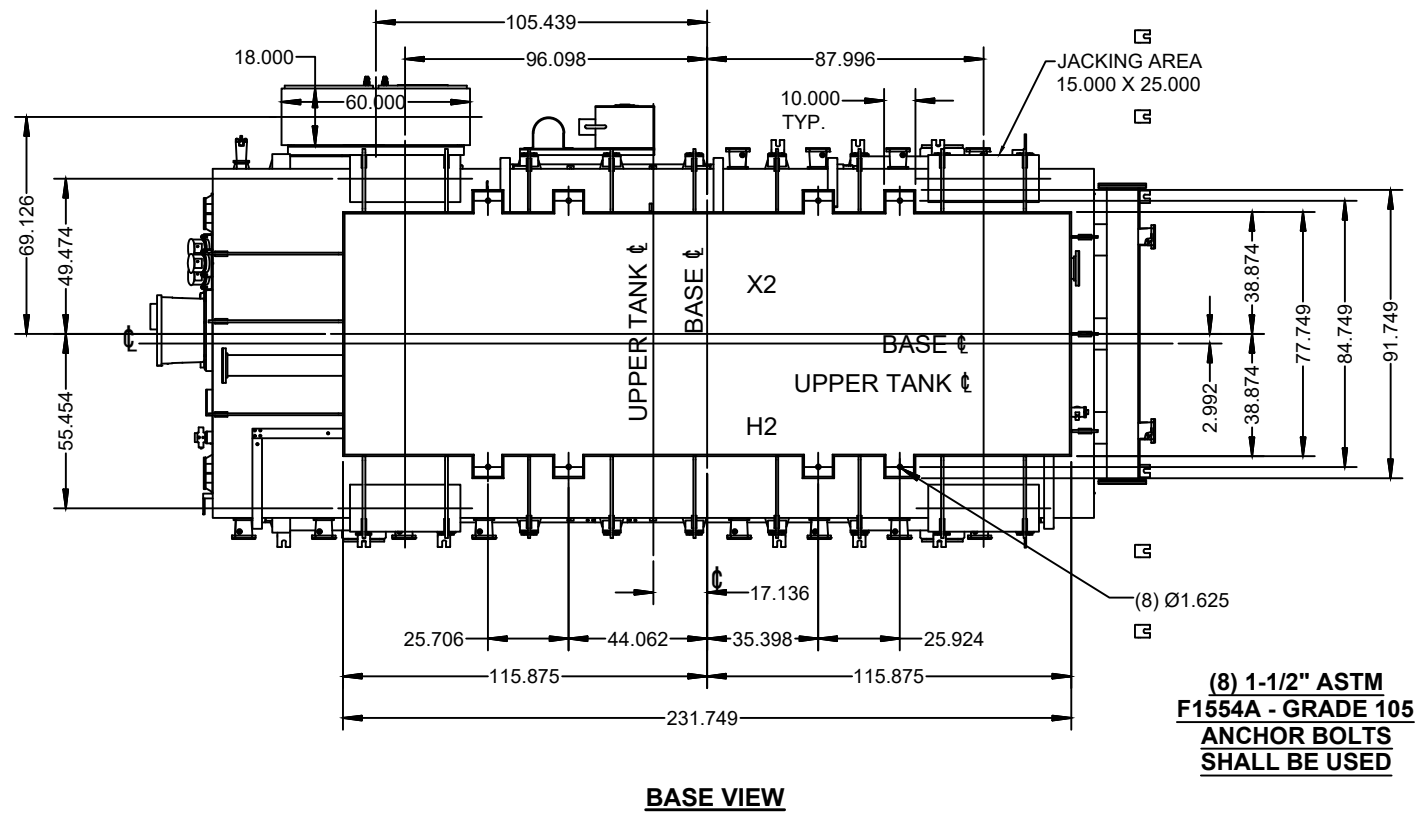
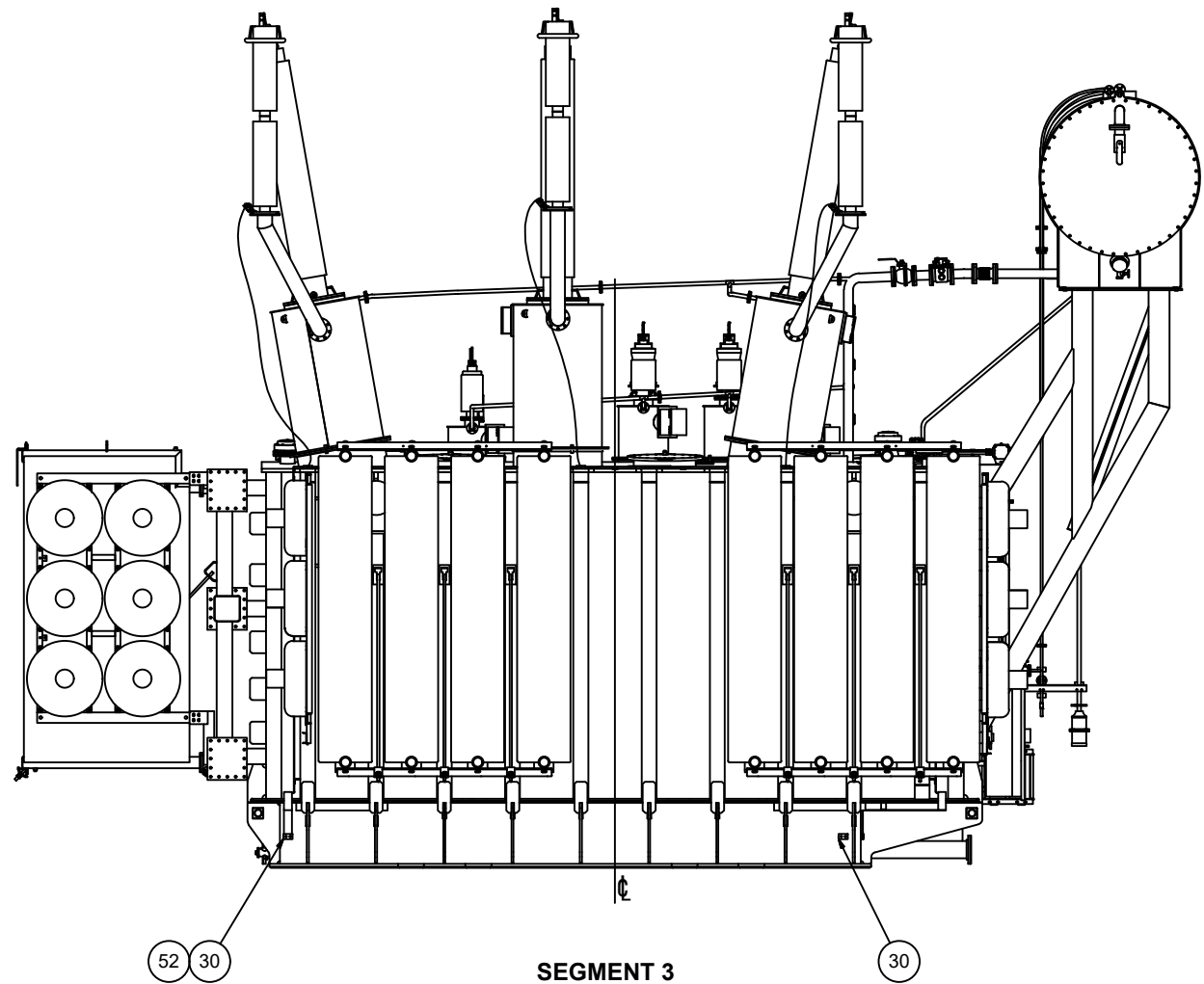
⊕ - CENTER OF GRAVITY

BOTTOM OIL



WEIGHTS AND OIL VOLUMES										SEGMENT 1												OTHER USES ARE PROHIBITED EXCEPT BY WRITTEN PERMISSION OF GTC.																																																																																																	
UNTANKING (UPPER BELL)		31765 Lbs.	14409 kg.	TOTAL OIL VOL @ 25C		118819 Lbs.	53895 kg.	CONSERVATOR OIL @ 25C		4228 Lbs.	1918 kg.	SURFACE		V05	PAINT	SITE		CUSTOMER		P.		TITLE: OUTLINE DRAWING		JOB #																																																																																															
TANK & FITTINGS		184298 Lbs.	83596 kg.			15908 Gal.	60218 Ltrs.			566 Gal.	2143 Ltrs.	EXTERIOR		V05	ANSI#70	TRANSFORMER		THRE-PHASE TRANSFORMER		ALL DIMENSIONS IN INCHES & IN mm IF SHOWN IN PARENTHESIS		DWN: JRM		CHK: JAJ		DATE: 11/17/2021																																																																																													
CORE & COILS		206637 Lbs.	93729 kg.	MAIN TANK OIL		98033 Lbs.	44921 kg.	LTC CONSERVATOR OIL @ 25C		306 Gal.	139 kg.	COVER		V05	NON-SKID	RATING @ 65°C RISE		126 / 168 / 210 MVA - ONAN / ONAF / ONAF																																																																																																					
INSULATION		18541 Lbs.	8410 kg.			13255 Gal.	50191 Ltrs.			41 Gal.	155 Ltrs.	INTERIOR		V01	WHITE	VOLTAGE		230 000 V ± 16 X 0.625% - 34 500 V - 13 800 V																																																																																																					
TOTAL WEIGHT		509754 Lbs.	231221 kg.	RADIATOR OIL (TOTAL)		14490 Lbs.	6573 kg.	LTC OIL		254 Gal.	115 kg.	UNDERBASE		V05	CARBOLINE 954	VECTOR GROUP		YNyn0+d1																																																																																																					
SHIPPING WITHOUT OIL		283063 Lbs.	128395 kg.			1940 Gal.	7344 Ltrs.			34 Gal.	129 Ltrs.																																																																																																												
8										7										6										5										4										3										2										1										D										49126MA002G										SHEET 101.0										REVISION 4									

LEGEND:
⊥ - CENTERLINE
⊕ - CENTER OF GRAVITY



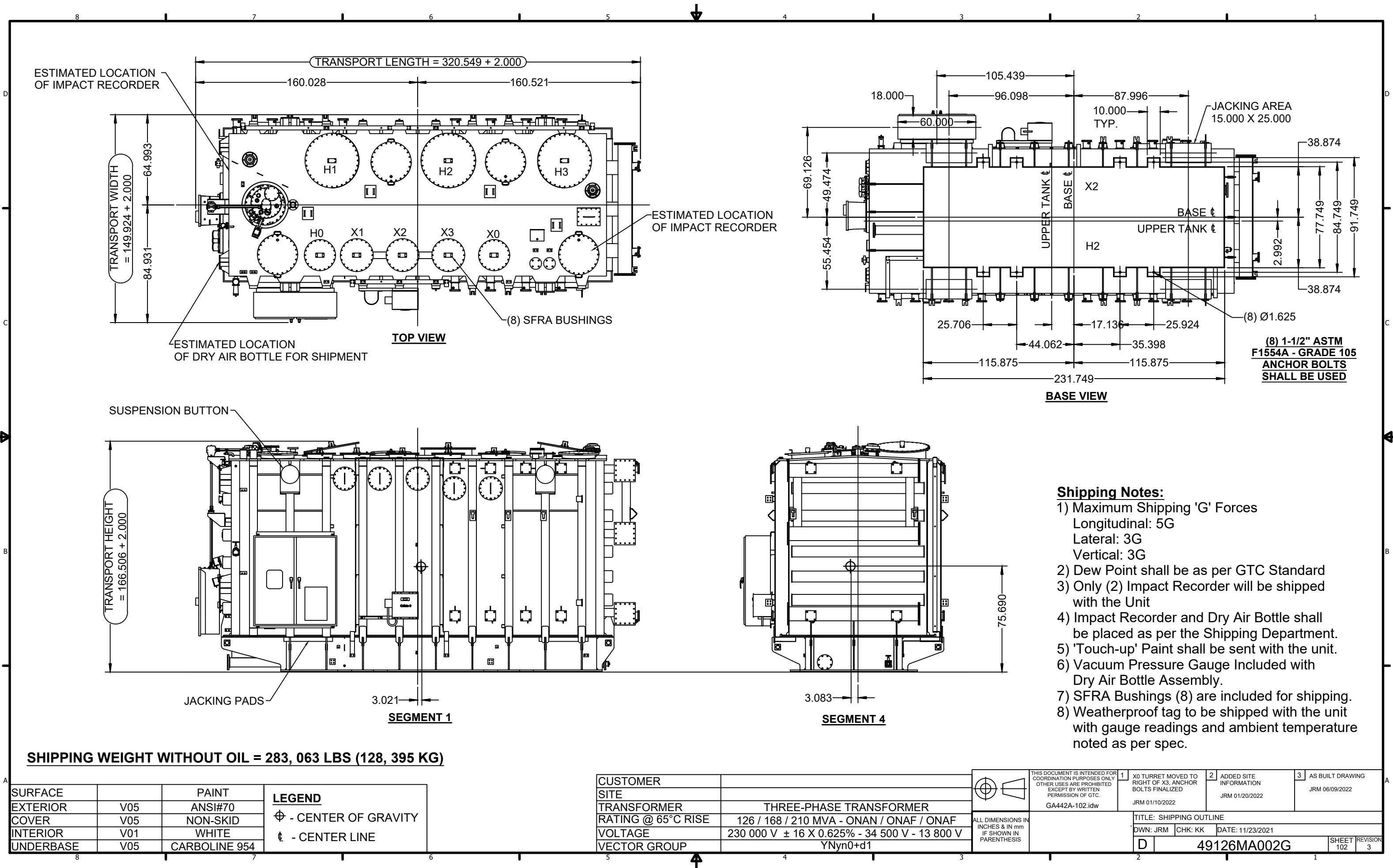
**(8) 1-1/2" ASTM
F1554A - GRADE 105
ANCHOR BOLTS
SHALL BE USED**

WEIGHTS AND OIL VOLUMES														
UNTANKING (UPPER BELL)	31765	Lbs.	14409	kg.	TOTAL OIL VOL @ 25C	118819	Lbs.	53895	kg.	CONSERVATOR OIL @ 25C	4228	Lbs.	1918	kg.
TANK & FITTINGS	184298	Lbs.	83596	kg.		15908	Gal.	60218	Ltrs.		566	Gal.	2143	Ltrs.
CORE & COILS	206637	Lbs.	93729	kg.	MAIN TANK OIL	99033	Lbs.	44921	kg.	LTC CONSERVATOR OIL @ 25C	306	Lbs.	139	kg.
INSULATION	18541	Lbs.	8410	kg.		13259	Gal.	50191	Ltrs.		41	Gal.	155	Ltrs.
TOTAL WEIGHT	509754	Lbs.	231221	kg.	RADIATOR OIL (TOTAL)	14490	Lbs.	6573	kg.	LTC OIL	254	Lbs.	115	kg.
SHIPPING WITHOUT OIL	283063	Lbs.	128395	kg.		1940	Gal.	7344	Ltrs.		34	Gal.	129	Ltrs.

SURFACE		PAINT
EXTERIOR	V05	ANSI#70
COVER	V05	NON-SKID
INTERIOR	V01	WHITE
UNDERBASE	V05	CARBOLINE 954

CUSTOMER	
SITE	
TRANSFORMER	THREE-PHASE TRANSFORMER
RATING @ 65°C RISE	126 / 168 / 210 MVA - ONAN / ONAF / ONAF
VOLTAGE	230 000 V ± 16 X 0.625% - 34 500 V - 13 800 V
VECTOR GROUP	YNyn0+d1

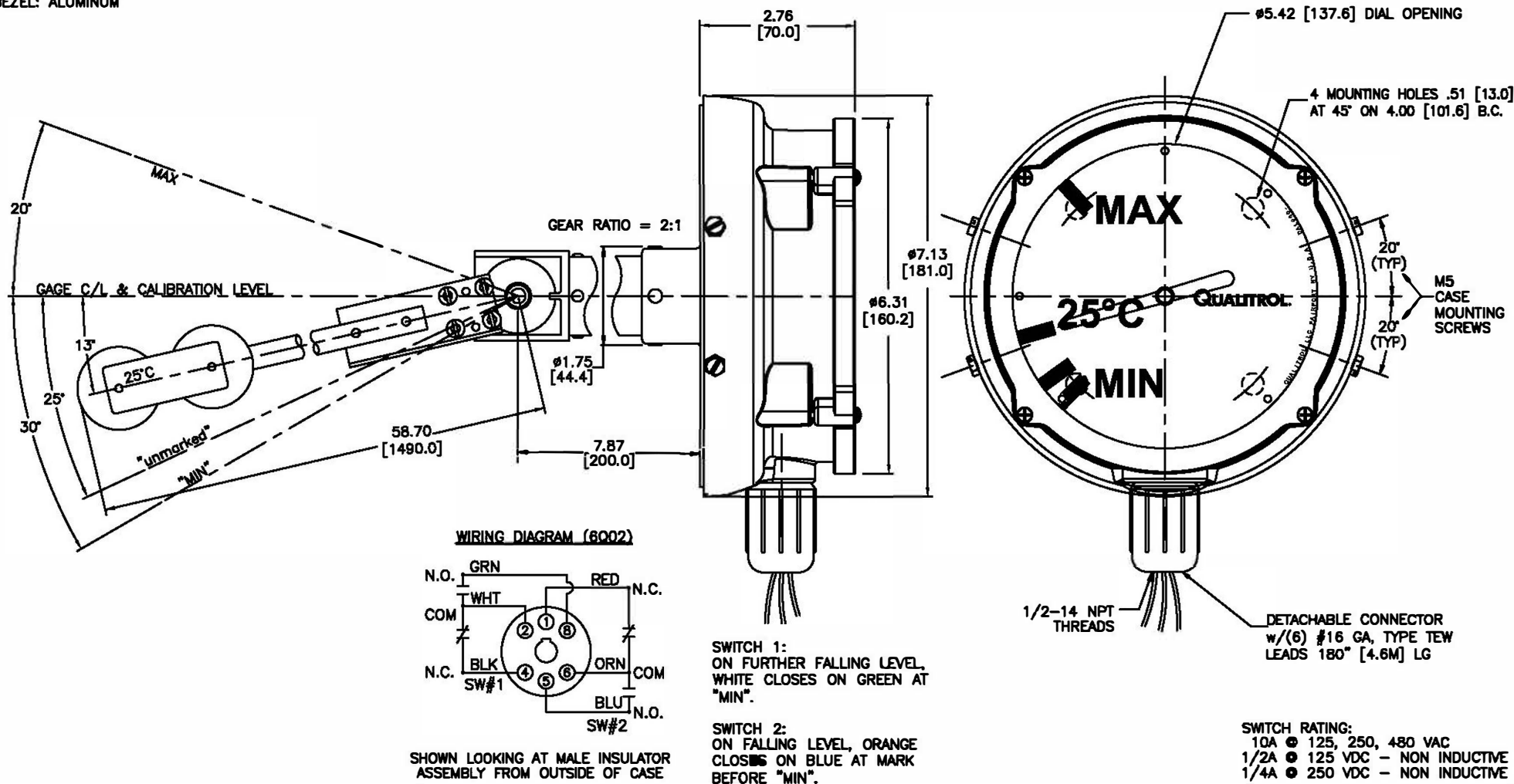
	THIS DOCUMENT IS INTENDED FOR COORDINATION PURPOSES ONLY OTHER USES ARE PROHIBITED EXCEPT BY WRITTEN PERMISSION OF GTC.	1	X0 TURRET MOVED TO RIGHT OF X3, ANCHOR BOLTS FINALIZED JRM 01/10/2022		2	ADDED SITE INFORMATION JRM 01/20/2022		3	AS BUILT DRAWING JRM 06/09/2022	
		TITLE: OUTLINE DRAWING								
		DWN: JRM		CHK: KK		DATE: 11/24/2021				
ALL DIMENSIONS IN INCHES & IN mm IF SHOWN IN PARENTHESIS		D		49126MA002G					SHEET 101.1	REVISION 3





FLANGE: ALUMINUM
SUPPORT TUBE: BRASS
BEZEL: ALUMINUM

GAGE MOUNTING ANGLE = 33 DEGREES DOWN FROM HORIZONTAL.



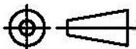
QUALITROL®

**DIMENSIONS AND TOLERANCES IN
ACCORDANCE WITH ASME Y14.5M-1994**

FORM 0-473 REV. F 04/06/06

6" LEVEL GAGE, VENTED

—	8-1-18	DRAWING CREATED	—
LET	DATE	ALTERATION	C.N.

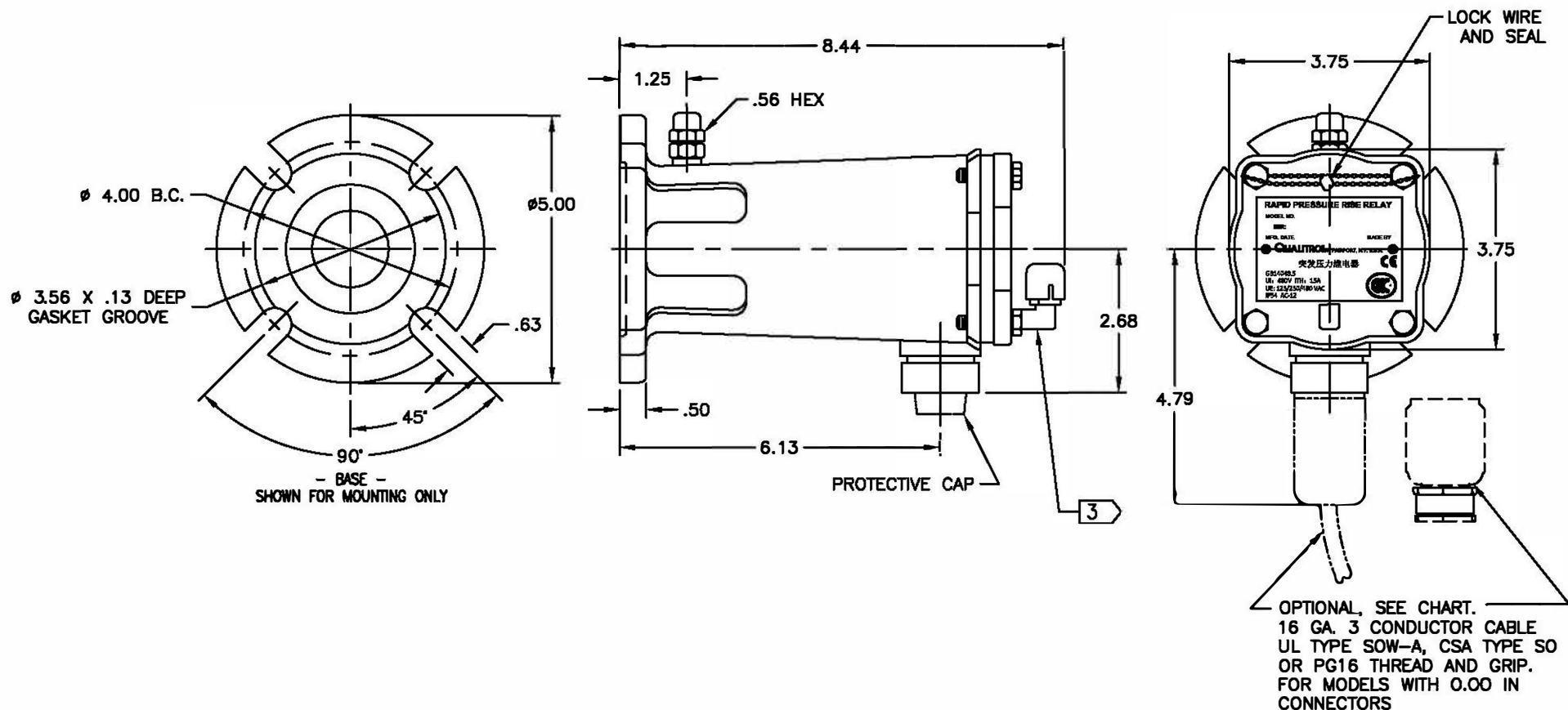


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NOTES:

1. MEETS QUALITROL PRODUCT PERFORMANCE SPECIFICATION PPS-9.
2. FOR OIL SPACE INSTALLATION ONLY

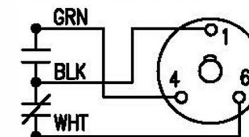
3. UNIT MUST BE MOUNTED HORIZONTALLY WITH VENT UP AS SHOWN.



SWITCH RATINGS

15 AMP @ 125, 250, 480 VAC
.5 AMP @ 125 VDC - NON-INDUCTIVE
.25 AMP @ 250 VDC - NON-INDUCTIVE

WIRING DIAGRAM



REVISION

MODEL NUMBER

LOW TEMP.
TEST

CONNECTOR
ASSEMBLY

CABLE
LENGTH

REMARKS

—	4-15-13	REV NAMEPLATE FOR CCC/CE	31153
—	5-25-10	ADDED ALTERNATE CONNECTOR VIEW	28146
LET	DATE	ALTERATION	C.N.

QUALITROL®

DIMENSIONS AND TOLERANCES IN
ACCORDANCE WITH ASME Y14.5M-1994

PH. (585) 586-1515
FAX (585) 377-0220

1385 FAIRPORT ROAD
FAIRPORT, NY 14450

FORM 0-473 REV. F 04/06/06

TITLE: RAPID PRESSURE RISE RELAY
(OIL)

DO NOT SCALE PRINT

DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH ASME Y14.5M-1994



NOTES:

1. P-V BLEEDER SETTINGS - WILL BE FURNISHED UNLESS OTHERWISE SPECIFIED.

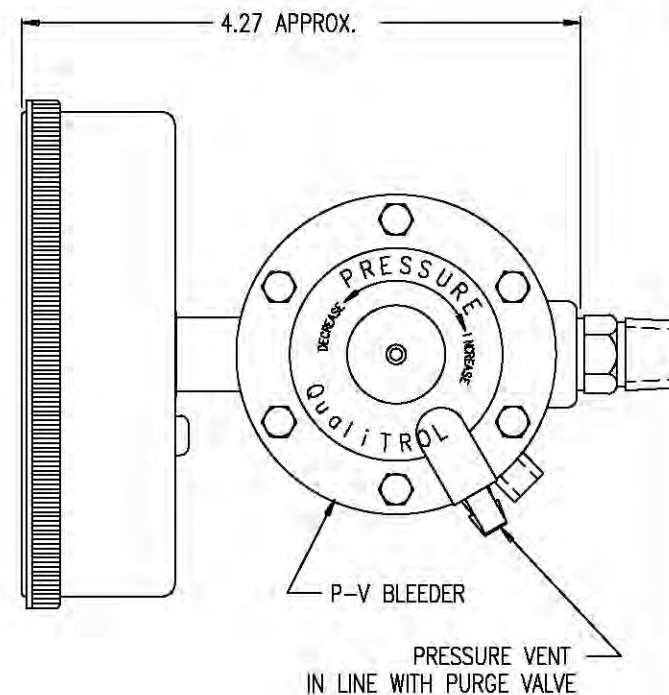
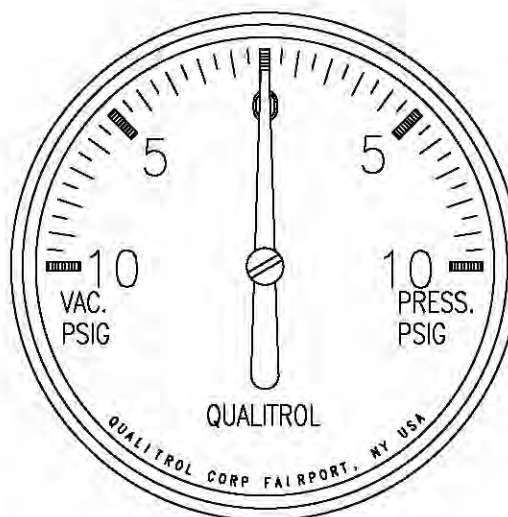
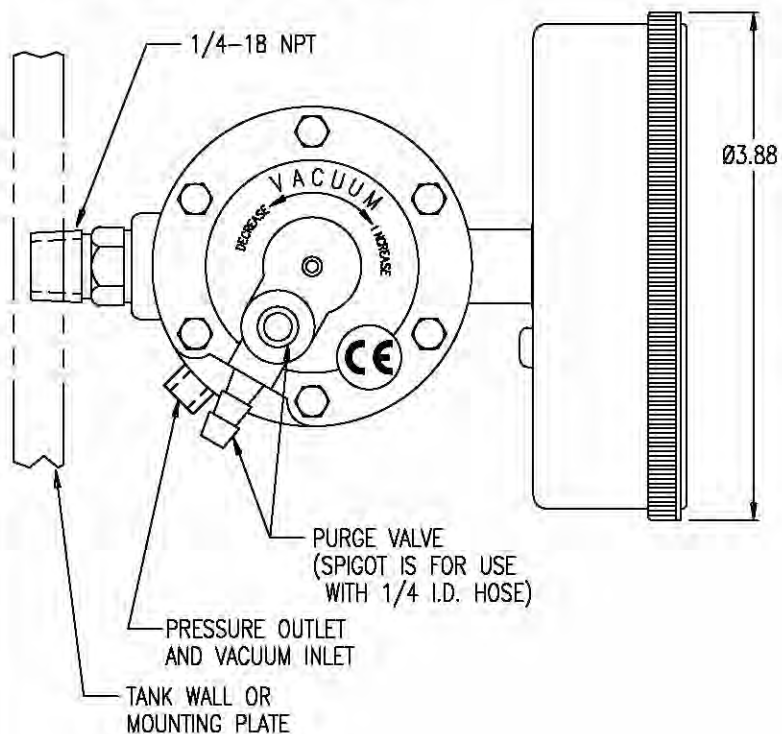
PRESSURE +9 P.S.I.

VACUUM -7 P.S.I.

2. DIAL COLORS:

BACKGROUND - FLAT BLACK

CHARACTERS AND POINTER - WHITE



NOTICE

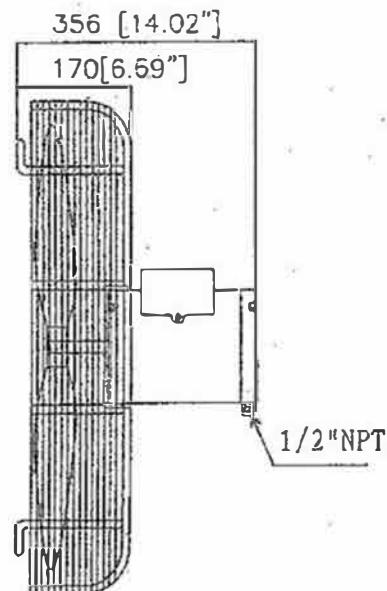
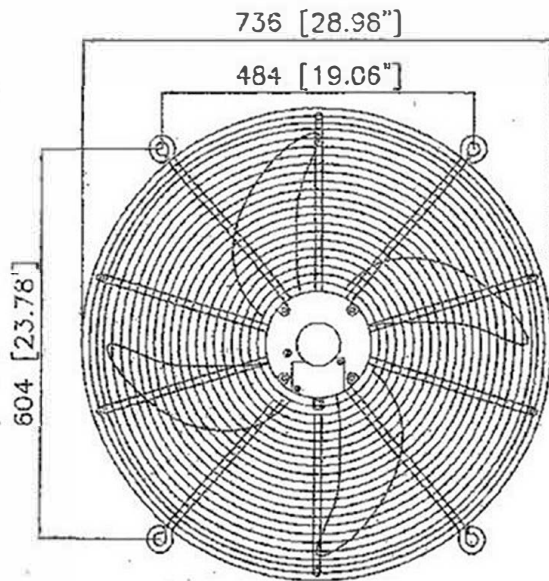
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TOLERANCE UNLESS NOTED,
[MM] IS FOR REF.:
2 PLC. DECIMAL $\pm .02$ [.5]
3 PLC. DECIMAL $\pm .005$ [.13]
ANGULAR ± 2 DEG.

FORM Q-487 REV. E 04/06/06
QUALITROL
FAIRPORT, NEW YORK

DRAWN	EAW	DATE	2/17-18
CHECK	-	-	-
APP'D	-	-	-

TITLE: P-V BLEEDER w/ GAGE	
MATERIAL:	
FINISH:	
SCALE 1:1	SHEET 1 OF 1 DRAWING NO.



unit: mm [in]

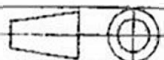
NOTES:

- 1: Noise is measured 6 feet from perimeter of the fan. act. to AMCA 300 "Reverberant Room Method for Sound Testing" . or ISO 13347 and BS 848 part 2. (tolerance ± 1.5 dBA)
- 2: Actual CFM measured act. to AMCA 210 "Caboratorg methods of testing fan for rating" and ISO 5801:1997 "Industrial fans-Performance testing using standardized airways".
- 3: Motor manufacture acc. to NEMA MG1 and IEC 60034. Finish paint of motor RAL 9016.
- 4: NPT 12 terminal compartment.
- 5: Guard: OSHA compliant.
- 6: Imprillers: 4 Blades. die-cast aluminum.

HERTZ	60		
HP	1/3		
PHASE	1		
VOLTS	104	115	125
R. P. M	1130	1146	1158
Input power	319 w	322 w	343 w
P. F	0.893	0.886	0.848
Starting	9.35	10.67	11.98
Nameplate		3.2	
Running	3.41	3.16	3.21
CFM	7496		
dBA	62		
WT kg	28		
FRAME	48		
Protection	IP 55 TENV		
Insulation	F		
AMB	55 C		
BB	Driver 6204 x 2		
BB	Non-driver 6203		

Customer's code:

No



Date

Name

Drawn 6-8-08

Design 6-8-08

Dppvd 6-8-08

Scale: 1:10 Page: 1 of 1

TRANSFORMER
COOLING FAN

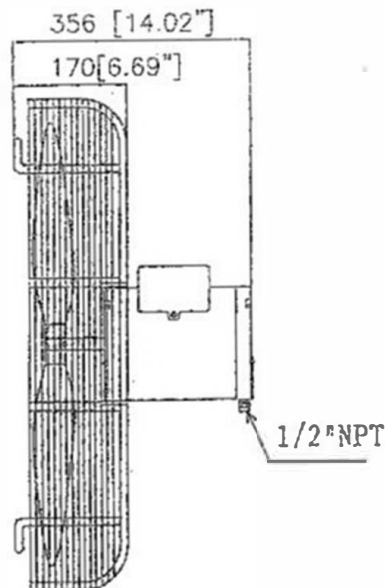
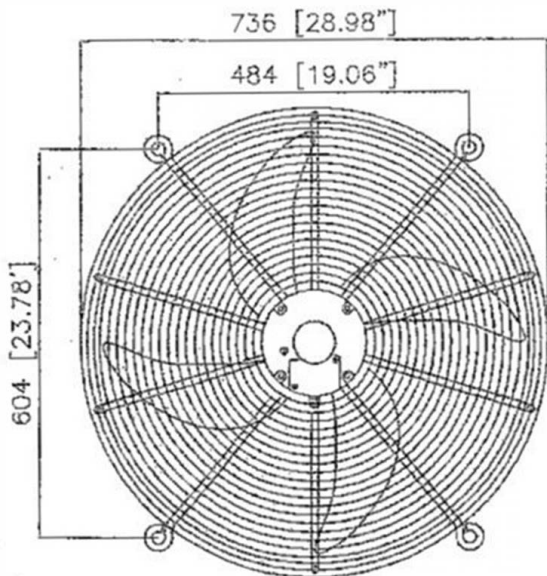
26" DIAMETER

A4

1/2

Purison Electric Component Co., Ltd.

Status Drawing on Date Name



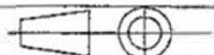
NOTES:

- 1: Noise is measured 6 feet from perimeter of the fan. act. to AMCA 300 "Reverberant Room Method for Sound Testing" or ISO 13347 and BS 848 part 2. (tolerance ± 1.5 dBA)
- 2: Actual CFM measured act. to AMCA 210 "Laboratory methods of testing fan for rating" and ISO 5801:1997 "Industrial fans-Performance testing using standardized airways".
- 3: Motor manufacture acc. to NEMA MG1 and IEC 60034. Finish paint of motor RAL 9016.
- 4: NPT 12 terminal compartment.
- 5: Guard: OSHA compliant.
- 6: Impellers: 4 Blades, die-cast aluminum.

AMPS	HERTZ	60		
	HP	1/3		
	PHASE	1		
	VOLTS	208	230	253
	R. P. M	1138	1152	1160
	Input power	313 w	328 w	345 w
	P. F	0.896	0.887	0.840
	Starting	4.89	5.5	6.39
	Nameplate	1.6		
	Running	1.66	1.58	1.62
MOTOR	CFM	7496		
	dBA	62		
	WT kg	28		
	FRAME	48		
	Protection	IP 55 TENV		
	Insulation	F		
	AMB	55 C		
	BB	Driver 6204 x 2		
	BB	Non-driver 6203		

Customer's code:

No



Date

Name

Drawn

6-8-08

甘虎峰

Design

6-8-08

8278

Dppvd

6-8-03

Scale: 1:10 Page: 1 of 1

TRANSFORMER
COOLING FAN

26" DIAMETER

A4

2/2

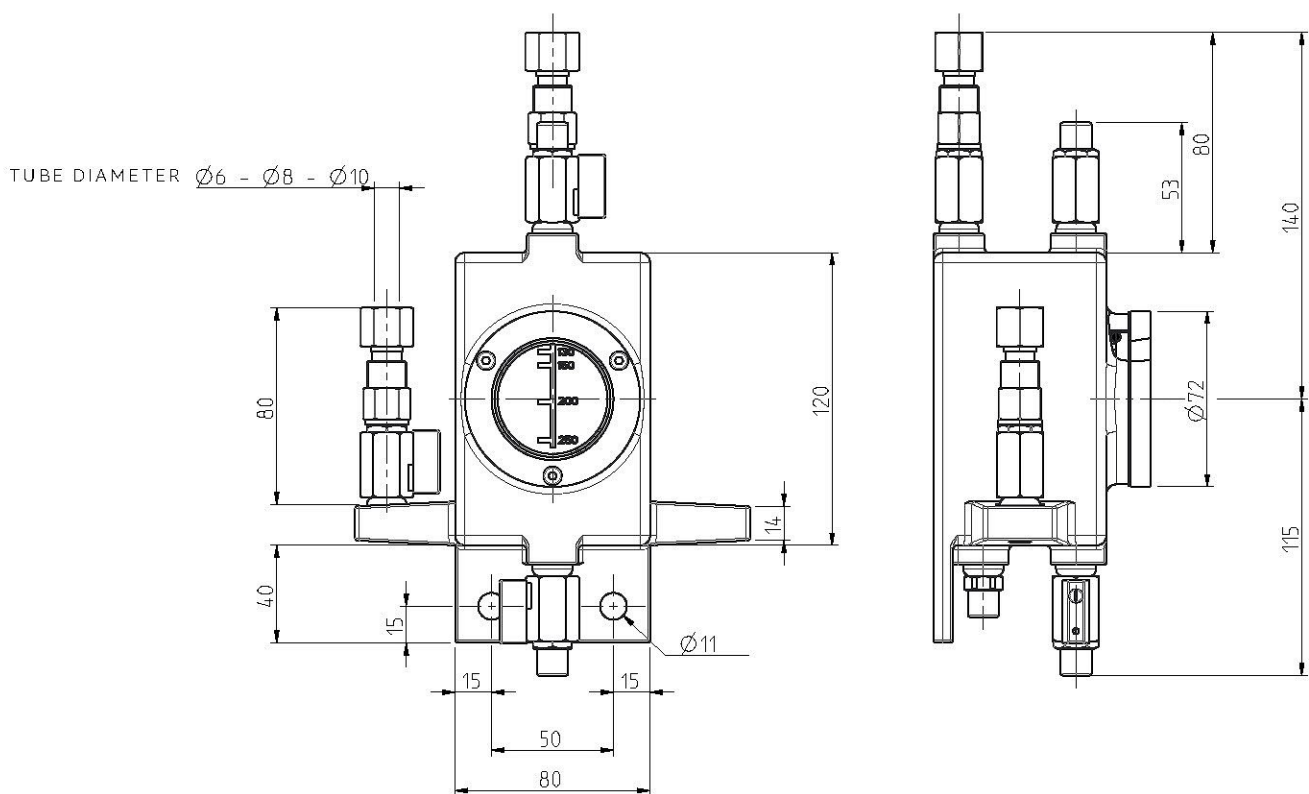
Furisen Electric Component Co., Ltd.

Status Drawing on Date Name

Gas sampling device Comem GSD

Technical data and dimensions

Comem GSD	Technical data
Material	
Housing	Aluminum casting, powder coated; Off shore model on request (surface treatment, not painted)
Characteristics data	
Installation	Indoors and outdoors, tropical proof
Ambient temperature	-40 to 80°C / -40 to 176°F
Oil temperature	-40 to 115°C / -40 to 239°F
Degree of protection	IP65 in accordance with EN60529
Nominal tube diameter	Ø6, Ø8 or Ø10 mm



Weight: 1.4 Kg

Gas sampling device Comem GSD

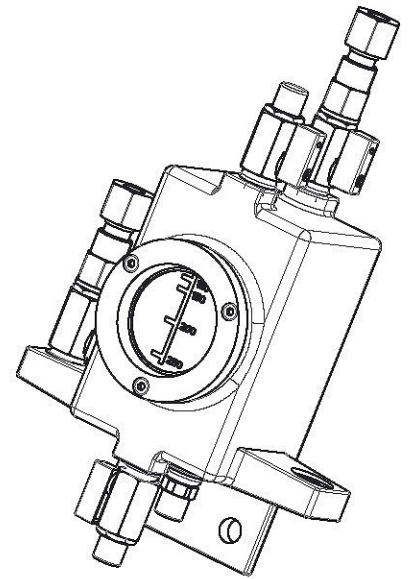
Order sheet

Date	
Rev.	
Customer reference	

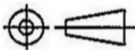
External Pipe Ø	
☐	10 mm
☐	8 mm
☐	6 mm

Gasket Material	
☐	NBR (from -40°C upto +120°C) Standard
☐	Fluorosilicone (from -60°C upto 200°C)

Corrosion protection class	
☐	C4 acc. to ISO 12944 (Standard)
☐	C5-M acc. to ISO 12944 (Not paintable)

**Notes:**

For different
configuration / options,
please contact ABB



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CASE: CAST ALUMINUM WITH ASA #70 LIGHT GRAY FINISH.

BEZEL: STAINLESS STEEL - REMOVABLE.
DIAL COLORS: BACKGROUND - BLACK
CHARACTERS - WHITE

POINTER COLOR: WHITE

MAX. INDICATING POINTER COLOR: RED

SWITCH SETTING INDICATOR COLOR: SILVER

LENS: POLYCARBONATE

SWITCH SETTING ADJUSTABILITY: 12°-108° C.

SWITCH DIFFERENTIAL: 7.5 °C ± 2.5 °C.

ACCURACY: ± 2% OF DIAL RANGE.

SWITCH RATING: UL/CSA RATING:

10 AMPS AND 1/3 HP @ 125, 250, VAC

1/2 AMP @ 125 VDC (NON-INDUCTIVE)

1/4 AMP @ 250 VDC (NON-INDUCTIVE)

4 AMPS @ 125 VAC "L" (LAMP LOAD)

DIELECTRIC STRENGTH: 1,500 VOLTS TO GROUND FOR 60 SECONDS.

TO ADJUST SWITCH INDICATOR(S):

A) REMOVE BEZEL AND LENS ASSEMBLY.

B) LOOSEN SCREWS AND MOVE SWITCH SETTING INDICATOR TO DESIRED SETTING. RE-TIGHTEN SCREWS.

C) REPLACE BEZEL AND LENS ASSEMBLY.

TO CHECK SWITCH OPERATION MANUALLY:

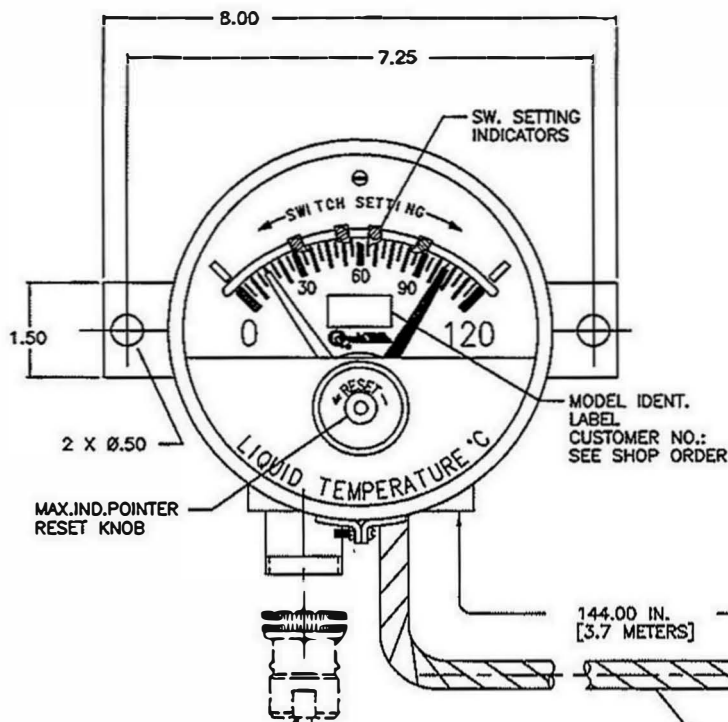
A) REMOVE BEZEL AND LENS ASSEMBLY.

B) TURN POINTER CLOCKWISE PAST THE SWITCH SETTING INDICATOR.

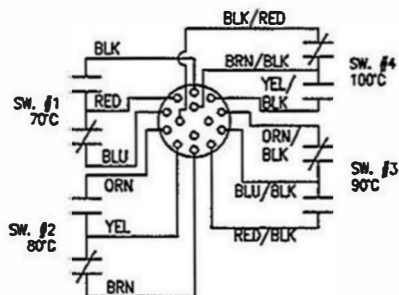
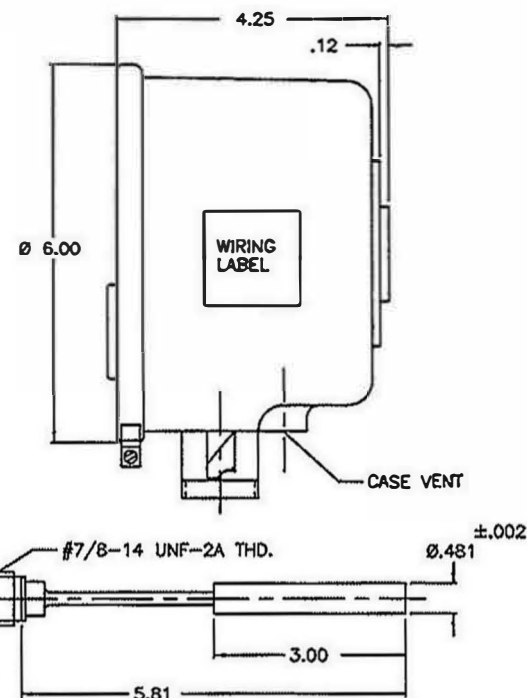
C) SLOWLY RETURN THE POINTER TO ITS ORIGINAL POSITION.

D) REPLACE BEZEL AND LENS ASSEMBLY.

CAUTION!! POINTER MUST NEVER BE TURNED COUNTERCLOCKWISE.



DETACHABLE CONNECTOR ASSY.
-NOT SUPPLIED WITH UNIT-



WIRING DIAGRAM-12A1

SHOWN LOOKING AT MALE INSUL. ASSY. FROM OUTSIDE OF CASE.

QUALITROL

DIMENSIONS AND TOLERANCES IN
ACCORDANCE WITH ASME Y14.5M-1994

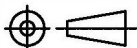
PH. (585) 586-1515
FAX (585) 377-0220

1385 FAIRPORT ROAD
FAIRPORT, NY 14450

FORM 0-473 REV. F 04/06/06

TITLE: 6" REMOTE MT. THERMO
(4 SWITCH) VENTED

LET	DATE	ALTERATION	C.N.
-	11-21-12	DRAWING CREATED	-

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CASE: CAST ALUMINUM WITH ASA #70 LIGHT GRAY FINISH.

BEZEL: STAINLESS STEEL - REMOVABLE.

DIAL COLORS: BACKGROUND - BLACK
CHARACTERS - WHITE

POINTER COLOR: WHITE

MAX. INDICATING POINTER COLOR: RED

SWITCH SETTING INDICATOR COLOR: SILVER

LENS: POLYCARBONATE

SWITCH SETTING ADJUSTABILITY: 18°-162°C.

SWITCH DIFFERENTIAL: 7.5 °C ± 2.5 °C.

ACCURACY: ± 2% OF DIAL RANGE.

SWITCH RATING: UL/CSA RATING:

10 AMPS AND 1/3 HP @ 125, 250, VAC

1/2 AMP @ 125 VDC (NON-INDUCTIVE)

1/4 AMP @ 250 VDC (NON-INDUCTIVE)

4 AMPS @ 125 VAC "L" (LAMP LOAD)

DIELECTRIC STRENGTH: 1,500 VOLTS TO GROUND FOR 60 SECONDS.

TO ADJUST SWITCH INDICATOR(S):

A) REMOVE BEZEL AND LENS ASSEMBLY.

B) LOOSEN SCREWS AND MOVE SWITCH SETTING INDICATOR TO DESIRED SETTING. RE-TIGHTEN SCREWS.

C) REPLACE BEZEL AND LENS ASSEMBLY.

TO CHECK SWITCH OPERATION MANUALLY:

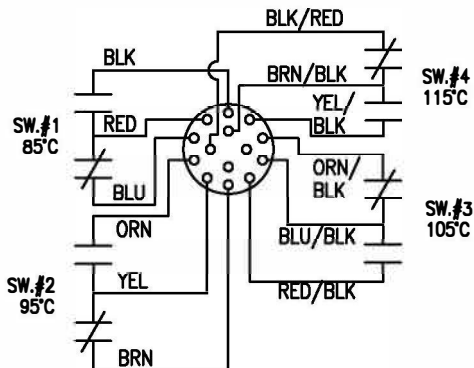
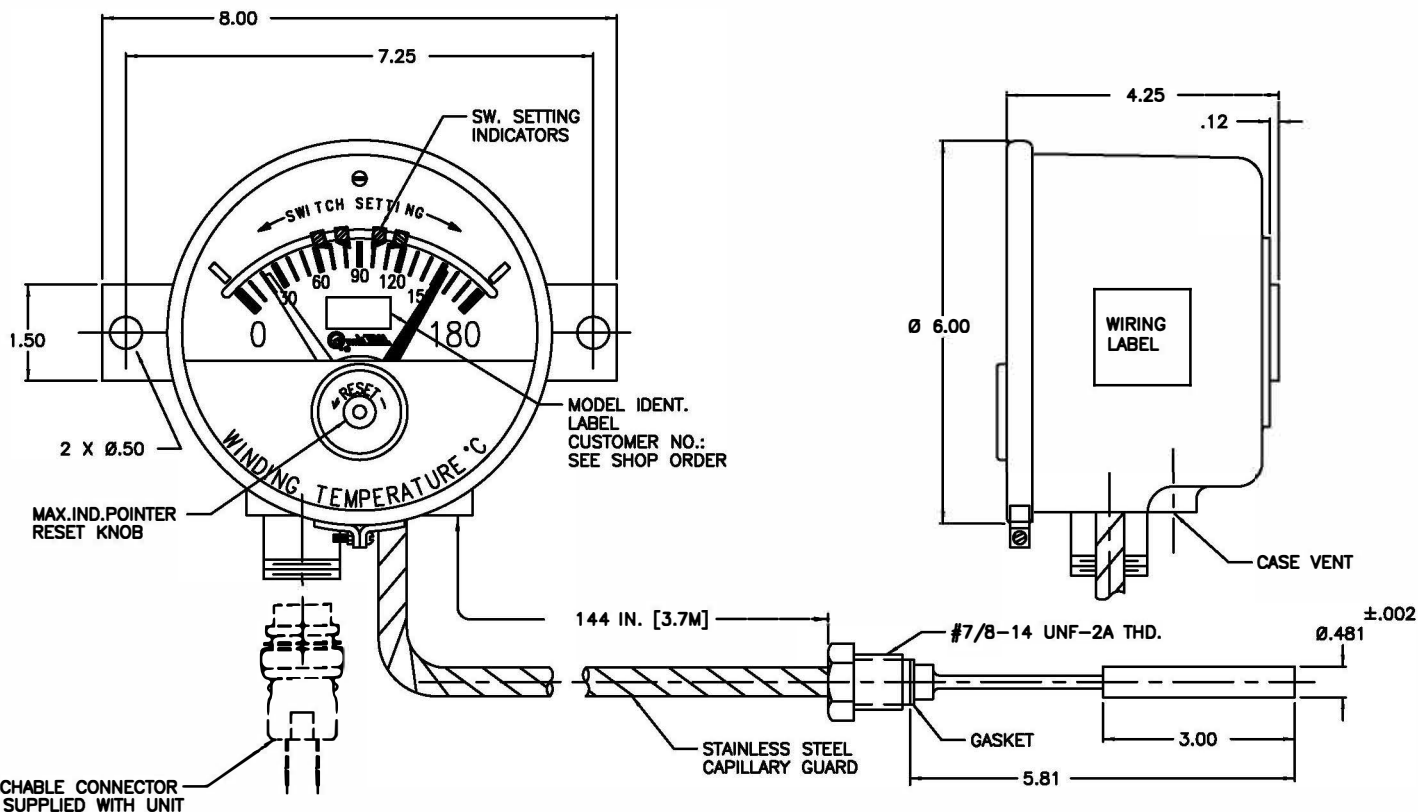
A) REMOVE BEZEL AND LENS ASSEMBLY.

B) TURN POINTER CLOCKWISE PAST THE SWITCH SETTING INDICATOR.

C) SLOWLY RETURN THE POINTER TO ITS ORIGINAL POSITION.

D) REPLACE BEZEL AND LENS ASSEMBLY.

CAUTION!! POINTER MUST NEVER BE TURNED COUNTERCLOCKWISE.



WIRING DIAGRAM-12A1

SHOWN LOOKING AT MALE INSUL. ASSY. FROM OUTSIDE OF CASE.

QUALITROL®

DIMENSIONS AND TOLERANCES IN
ACCORDANCE WITH ASME Y14.5M-1994

PH. (585) 586-1515
FAX (585) 377-0220

1385 FAIRPORT ROAD
FAIRPORT, NY 14450

FORM Q-473 REV. F 04/06/06

TITLE: 6" REMOTE MT. THERMO
(4 SWITCH) VENTED

—	9-19-13	DRAWING CREATED	—
LET	DATE	ALTERATION	C.N.

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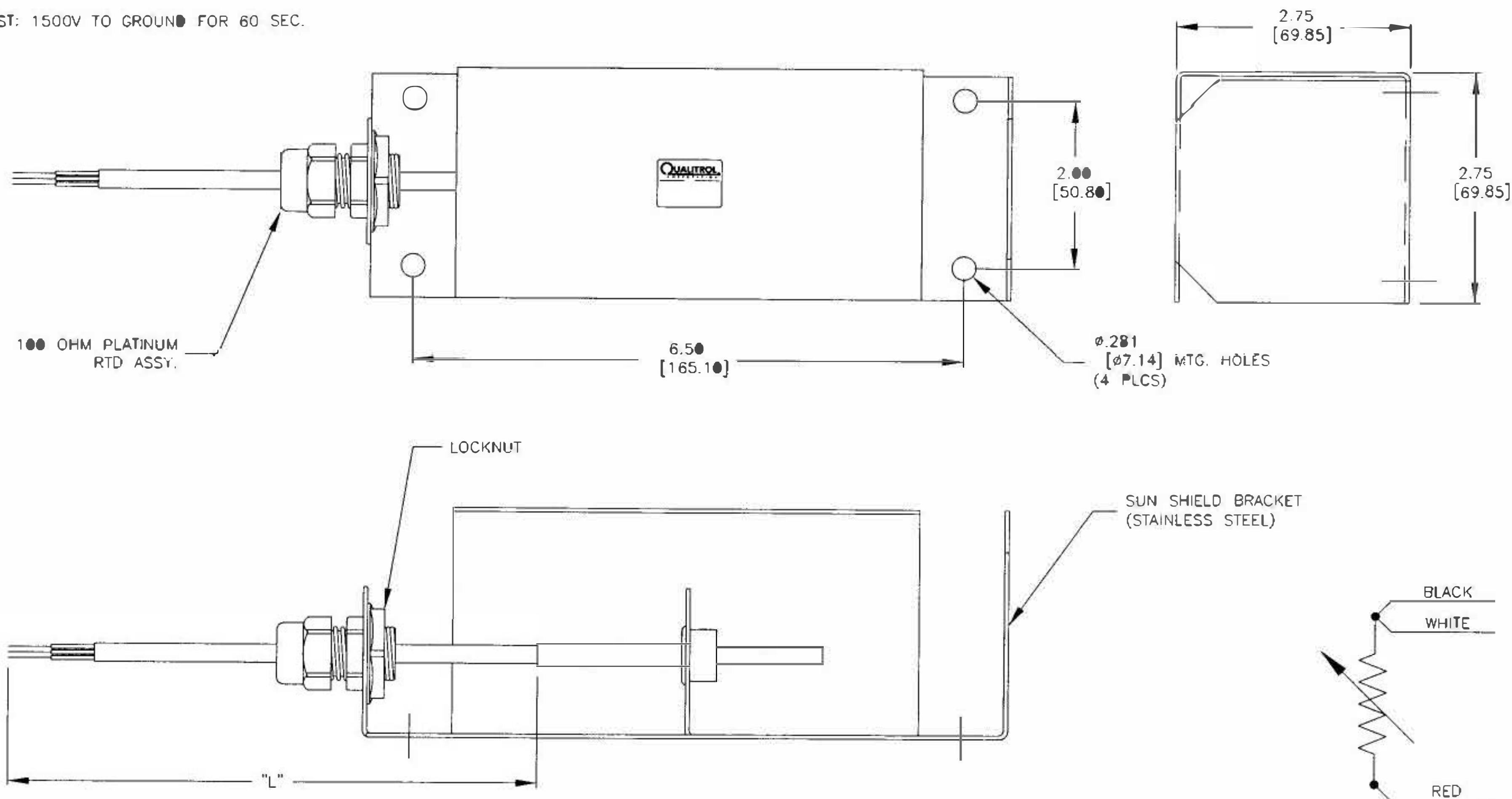
ALL DIMENSIONS ARE FOR REFERENCE ONLY.
DIMENSIONS ARE SHOWN IN INCHES [MILLIMETERS].

CABLE LENGTH "L"
144.00"

CABLE:
CONDUCTOR => 22 AWC, 3 CONDUCTOR SHIELDED
CONDUCTOR INSULATION => PTFE
JACKET INSULATION => PTFE, -55°C TO +200°C
BRAID => SILVER PLATED COPPER BRAID 85% COVERAGE
NOMINAL SIZE => #149

RTD: 100 OHM AT 0°C
EUROPEAN 'E' CURVE .00385 ohms/ohms/°C

INSULATION TEST: 1500V TO GROUND FOR 60 SEC.



-	5-20-03	ADDED PTS. - 04 THRU - 11	23849
-	1-21-02	@RTD: 100 OHMS@0°C WAS 100 OHMS@25°C	22720
-	3-8-01	RELEASED	21970
LET	DATE	ALTERATION	C.N.

QUALITROL
CORPORATION
DIMENSIONS AND TOLERANCES IN
ACCORDANCE WITH ASME Y14.5M-1994

PH. (585) 586-1515
FAX (585) 377-0220
1385 FAIRPORT ROAD
FAIRPORT, NY 14450
FORM Q-473 REV.D 10/22/02

TITLE: PLATINUM RTD - AMBIENT
TEMPERATURE SENSOR

RATINGS (kV rms)		TOV (kV rms)	MAXIMUM RESIDUAL VOLTAGE WITH CURRENT WAVE (kV PEAK)				
VOLTAGE V _r	MCOV V _{mco v}	WITH PRIOR ENERGY SINGLE IMPULSE OF 4.5 kJ/kVr	SPL (SIPL) 30/60 μS	LPL (LIPL) 8/20 μS			FOW 0.5 μS
			500 A	3 kA	10 kA	20 kA	10 kA
STANDARD CREEP							
36	29.0	Ø10 sec = 39.4	71.3	80.0	86.4	95.5	94.2

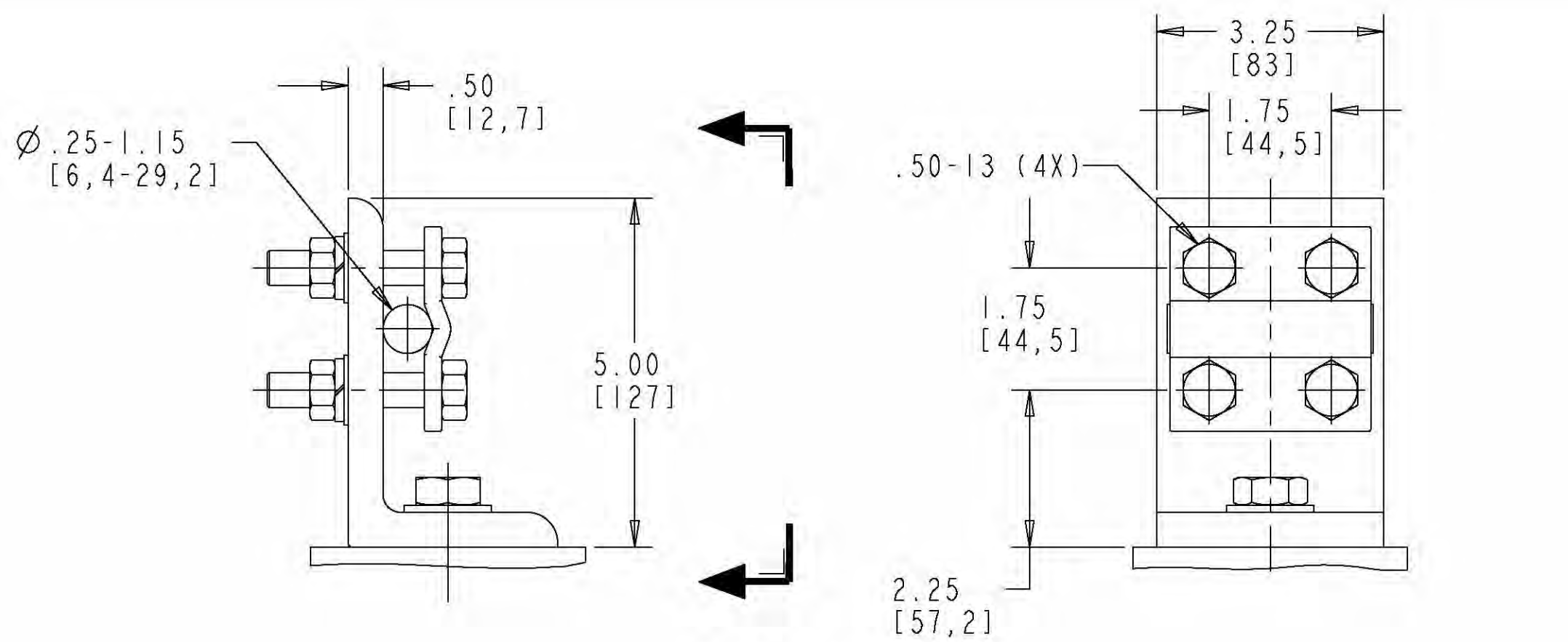
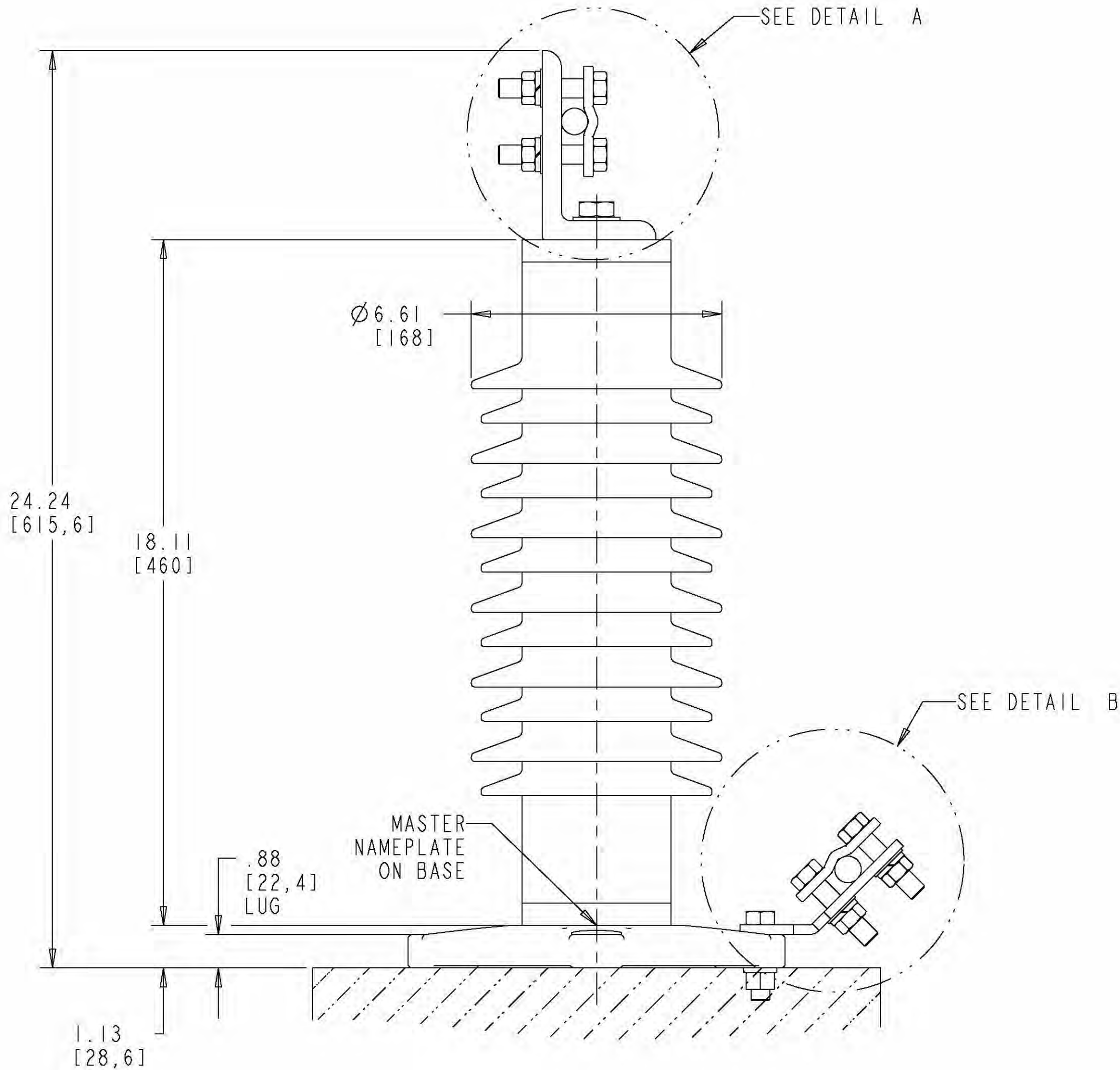
ABB POLIM-S SURGE ARRESTER					
STYLE NO.	*	SERIAL NO.	*		
RATING	*	kV RMS	MCOV	*	kV RMS
WT.	33 lbs	MFG DATE	*		
GRADING RING	NONE	PRESS. RELIEF CLASS	65	kA RMS	
MADE IN USA		IEEE / ENERGY / IEC CLASS	STATION / E / 3		

MASTER NAMEPLATE

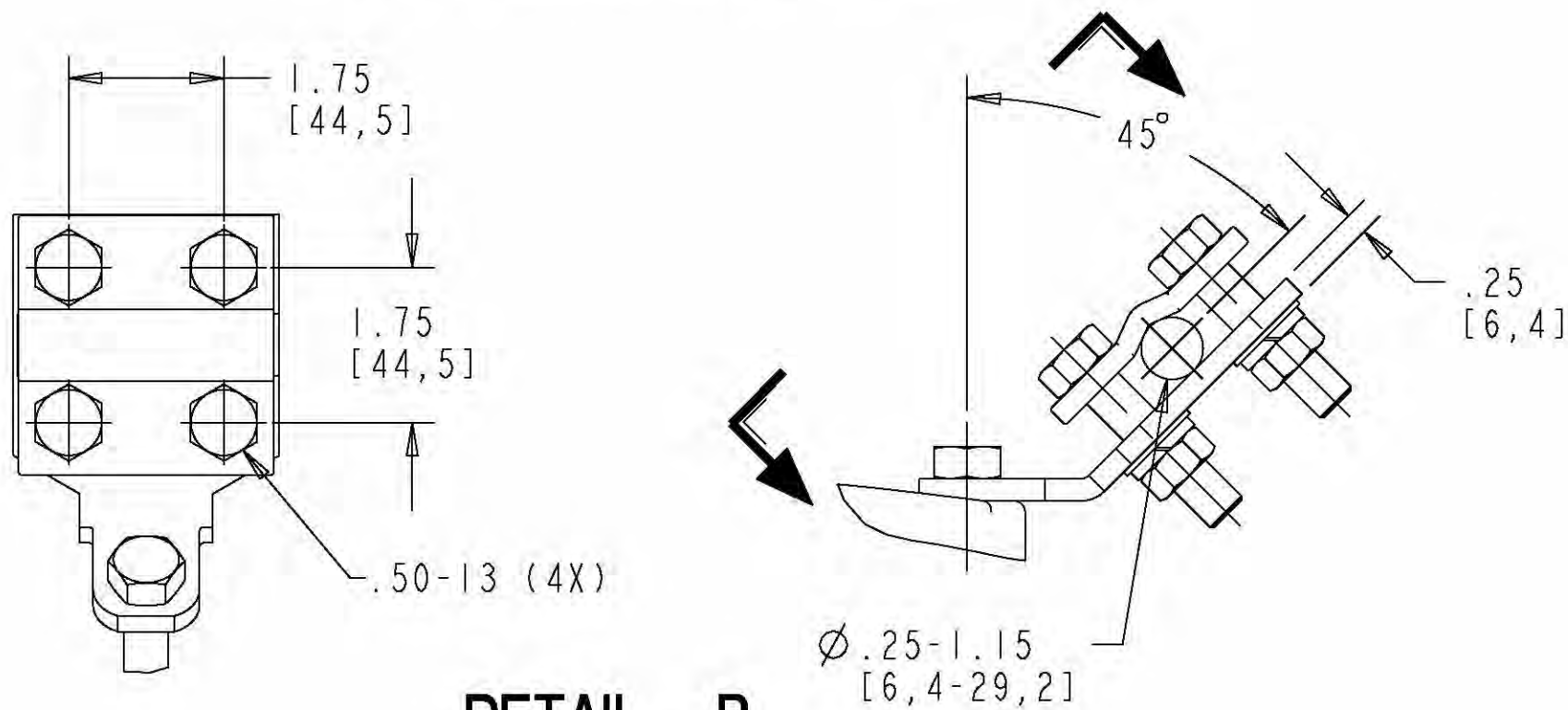
* ASSIGNED AT THE TIME OF PRODUCTION AND
RECORDED IN THE PROPER LOCATIONS

NOTES:

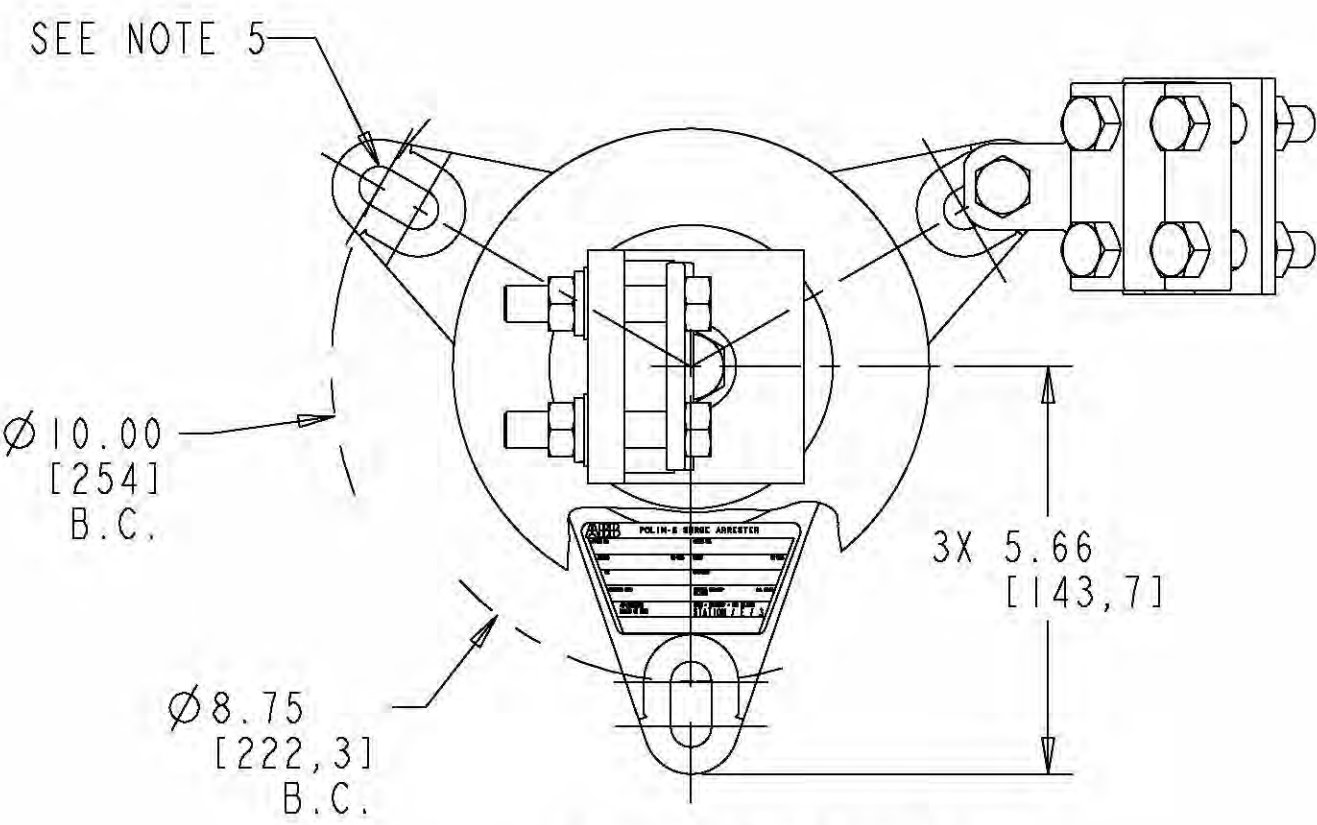
- FOR MINIMUM PHASE-TO-GROUND AND PHASE-TO-PHASE CLEARANCES REFER TO IEEE C62.11 AND C62.22. OTHER APPARATUS STANDARDS, SPECIFICATIONS OR LOCAL CODES MAY SUPERCEDE DISTANCES SPECIFIED.
- ARRESTER ASSEMBLY CONSISTS OF ARRESTER UNIT, LINE & GROUND TERMINALS WITH CLAMPS, AND .50 INCH HARDWARE.
- LINE & GROUND TERMINALS CAN ACCOMODATE CU OR AL CABLE SIZE NO. 2 TO 1000 MCM (.25 [6.4] TO 1.15 [29.2]) DIA.
- LINE & GROUND TERMINAL CLAMP CAN BE MOUNTED AS SHOWN OR ROTATED 90° AS REQUIRED.
- MTG. SLOTS FOR .50 INCH HARDWARE (3) EQUALLY SPACED, THICKNESS OF LUG = .88 [22.4]
- DESIGNED FOR VERTICAL OR HORIZONTAL MOUNT.
- MAX CANTILEVER LOAD 20,000 in-lbs [2260 Nm].
- PRESSURE RELIEF CLASS IS 65 kA RMS SYMMETRICAL.
- NOMINAL CREEPAGE DISTANCE 42.2 [1073].
- MAXIMUM ARRESTER WEIGHT 33.0 lbs [15.0 kg].



DETAIL A
(ENLARGED)
SCALE 0.500
LINE TERMINAL (ALUMINUM)
(CAN BE ADJUSTED TO ANY ANGLE)



DETAIL B
(ENLARGED)
SCALE 0.500
GROUND TERMINAL (GALVANIZED)
(CAN BE ATTACHED TO ANY LUG)



DRILLING PLAN

ALL DIMENSIONS IN INCHES [MM] UNLESS NOTED

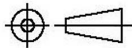
This drawing is the property of ABB INC. and contains
proprietary and confidential information which must not be duplicated,
used or disclosed other than expressly authorized by ABB INC.

POLIM-S / IEEE STATION E / IEC CLASS 3
POLYMER ARRESTER
12 SHED HOUSING OUTLINE

ABB

ABB Inc.
Power Products Division
100 Distribution Circle, Mt Pleasant, PA 15666

CAD DRW PRO/E WILDFIRE 2.0		SHEET 1 OF 1	
SCALE 0.20			
ASSY. DWG.			REV.
DWN	KRYGOWSKI		
CHKD	DJK		
APPD	MGM		



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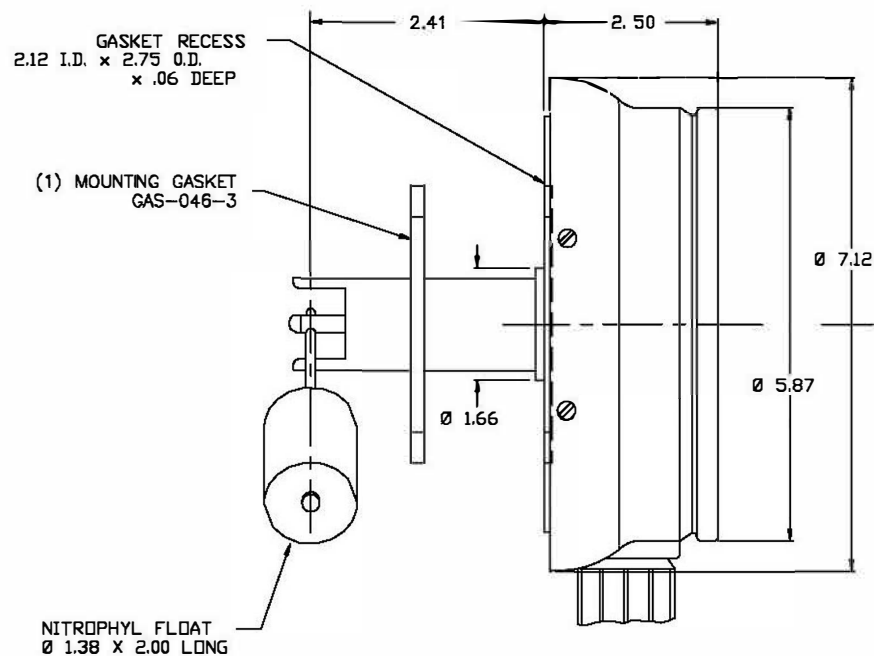
ALL DIMENSIONS FOR REFERENCE ONLY.

CASE ASSEMBLY: SEALED AGAINST DUST AND MOISTURE
DIAL: WHITE CHARACTERS, BLACK BACKGROUND
POINTER: WHITE

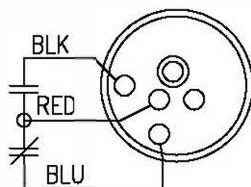
LENS: GLASS
CASE: ALUMINUM, PAINTED ASA 70 LIGHT GRAY EPOXY
FLANGE: ALUMINUM
SUPPORT TUBE: BRASS
FLOAT ROD: BRASS

ALL ANGULAR DEGREES SHOWN REFLECT FLOAT ROD TRAVEL.
FLOAT ROD: MUST EXTEND FROM COUNTERBORED SIDE OF SHAFT
FLOAT LEADS POINTER BY .0°

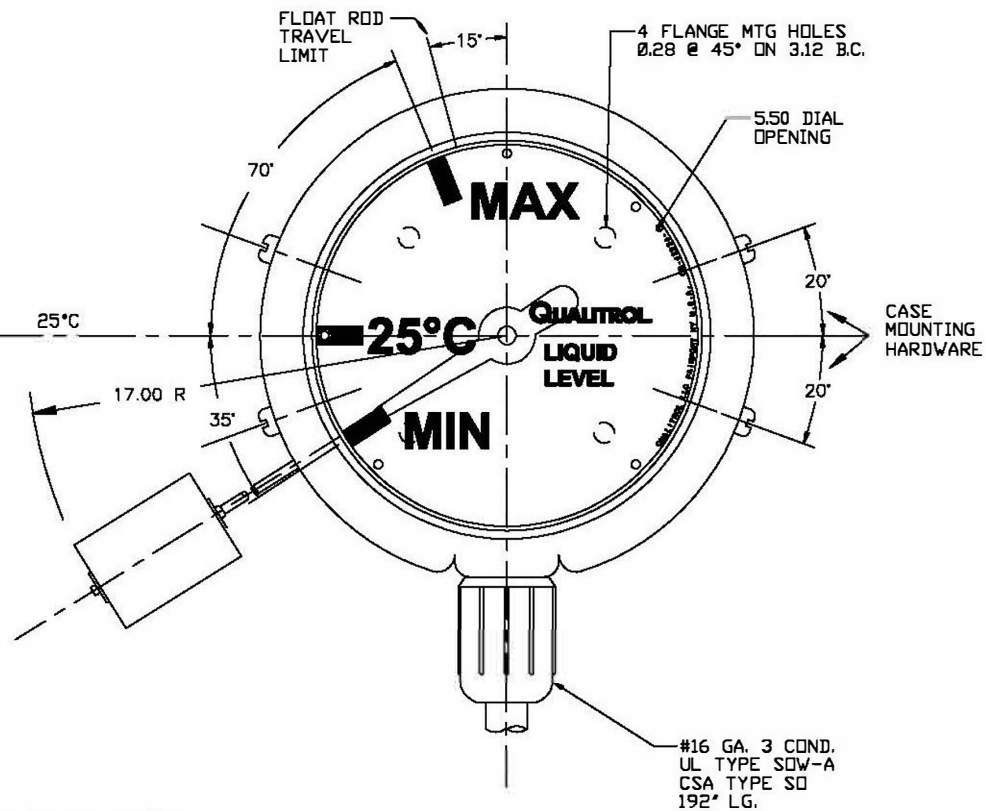
INSULATION TEST: 2000V TO GROUND FOR 60 SECONDS
SWITCH RATING:
10A @ 125, 250, 480 VAC
1/2A @ 125 VDC } NON INDUCTIVE
1/4A @ 250 VDC }



WIRING DIAGRAM- ANSI



ON FALLING LEVEL,
RED CLOSSES ON
BLACK AT MIN



PRELIMINARY
DRAWING

SHOWN LOOKING AT MALE INSULATOR ASSEMBLY FROM OUTSIDE OF CASE

LET	DATE	ALTERATION	C.N.
	4/7/14	PRELIMINARY	P32402

QUALITROL

DIMENSIONS AND TOLERANCES IN
ACCORDANCE WITH ASME Y14.5M-1994

PH. (585) 586-1515
FAX (585) 377-0220

1385 FAIRPORT ROAD
FAIRPORT, NY 14450
FORM 0-473 REV. F 04/06/06

TITLE:

6" LEVEL GAGE

CURRENT RATING

400 AMP DRAW LEAD CONNECTION
1200 AMP BOTTOM CONNECTION

LT. GRAY

DATE 3-25-03 REV. DATE 11-11-08

MAXIMUM ALTITUDE 10000 ft [3000 m]

NET WEIGHT 90 lb [41 kg]

ANGLE OF INCLINATION FROM THE VERTICAL 0° TO 60°

MIN. CREEP 30.00 [762.0] MIN. STRIKE 14.00 [355.6]

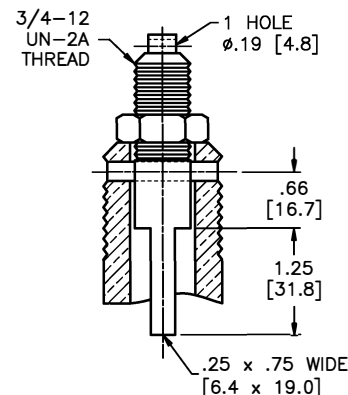
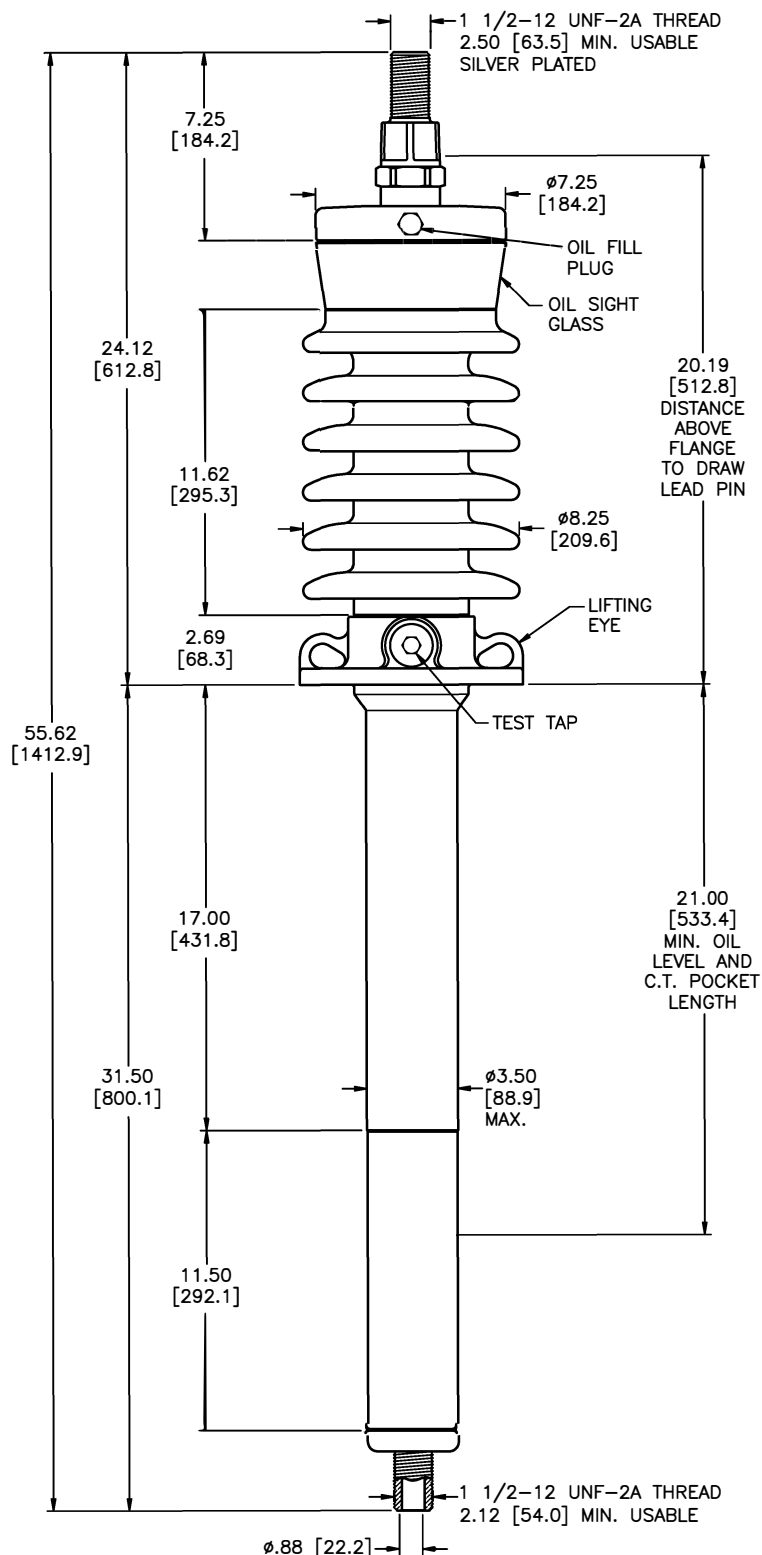
CREEP/U_y 1.36 [34.6] STD. IEEE C57.19.00/01

PCB CONTENT OF OIL IS LESS THAN 1 ppm

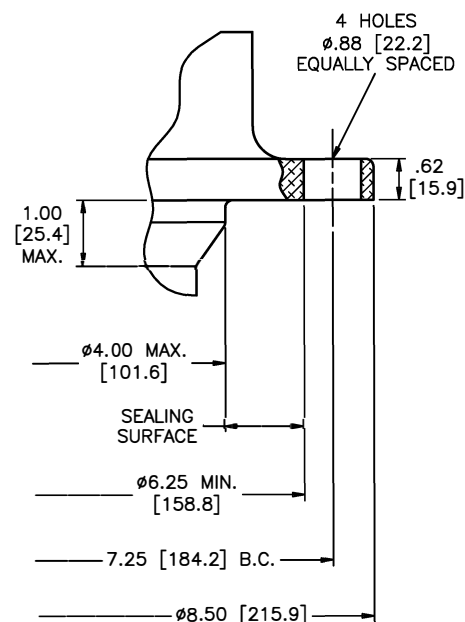
TITLE

POC BUSHING

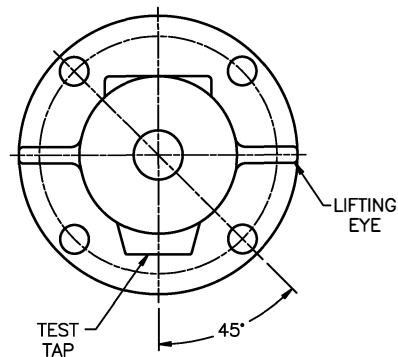
NOTE: ACTUAL DRAW-LEAD CURRENT
RATING IS DETERMINED BY
CABLE SIZE



DRAW LEAD ASSEMBLY



MOUNTING FLANGE DETAIL

FLANGE ALIGNMENT DIAGRAM
TOP VIEW

DIMENSIONS IN INCHES [MILLIMETERS]

REVISION NEW BOTT. CAP & LOW. PORCELAIN, MAX. ALTITUDE WAS 3300 ft, REMOVED °C, TEMPERATURE GOVERNED BY IEEE STANDARDS, WEIGHT WAS 111 lb

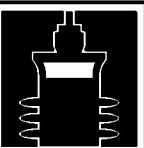
SCALE 1/5

NOTE TO PURCHASER:
ALL SALES OF PCORE PRODUCTS ARE SUBJECT
TO OUR STANDARD TERMS AND CONDITIONS
AND THE LIMITED WARRANTIES THEREUNDER

WWW.PCOREELECTRIC.COM

PCORE

ELECTRIC



DATE6-8-73

REV. DATE11-11-08

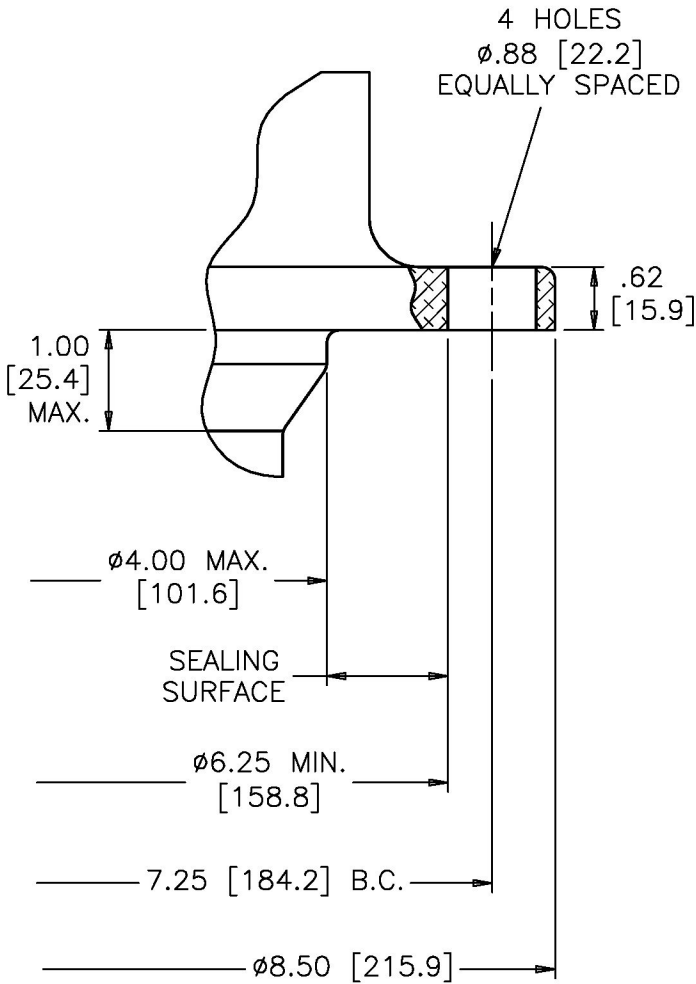
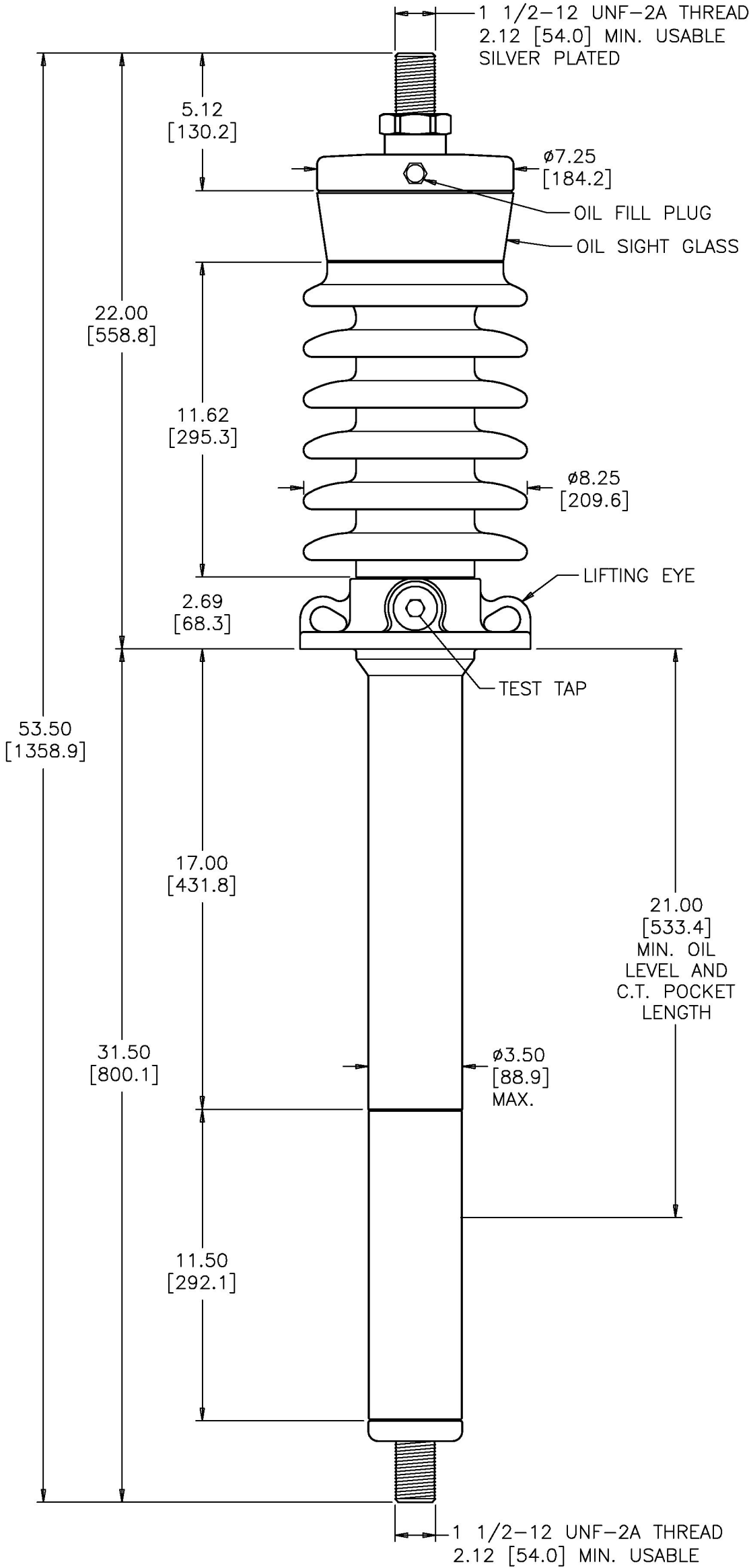
MAXIMUM ALTITUDE10000 ft [3000 m]

NET WEIGHT111 lb [50 kg]

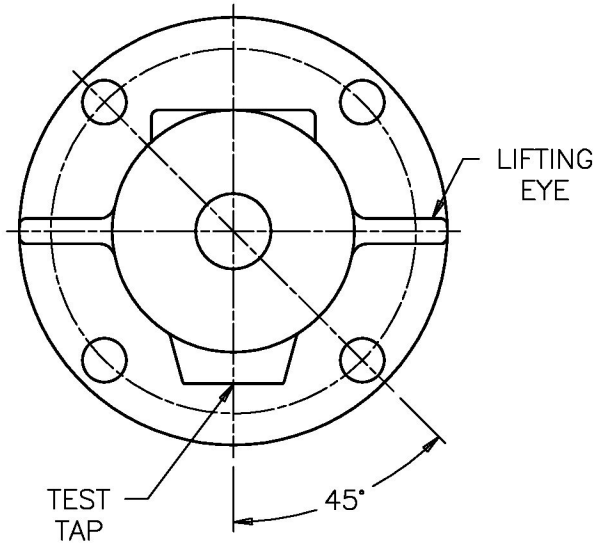
ANGLE OF INCLINATION FROM THE VERTICAL0° TO 60°

KV CLASS	34.5 kV	BIL	200 kV	Uy	22 kV
CURRENT RATING					
2000 AMP BOTTOM CONNECTION					
MIN. CREEP	30.00 [762.0]	MIN. STRIKE	14.00 [355.6]		
CREEP/Uy	1.36 [34.6]	STD.	IEEE C57.19.00/01		
PCB CONTENT OF OIL IS LESS THAN 1 ppm					

DWG.		GLAZE COLOR
SD	VTC PN# 55724568	LT. GRAY
TITLE		
POC BUSHING		



MOUNTING FLANGE DETAIL



FLANGE ALIGNMENT DIAGRAM
TOP VIEW

DIMENSIONS IN INCHES [MILLIMETERS]

REVISION	NEW BOTT. CAP & LOW. PORCELAIN, MAX. ALTITUDE WAS 3300 ft, REMOVED °C, TEMPERATURE GOVERNED BY IEEE STANDARDS	SCALE	1/5
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PCORE
ELECTRIC

WV CLASS 34.5 kV BIL 200 kV U_y 22 kV

CURRENT RATING

4000 AMP BOTTOM CONNECTED

DRG. **GLAZE COLOR**

SD LT. GRAY

DATE 2-27-08 REV. DATE 5-2-16

MAXIMUM ALTITUDE 10000 ft [3000 m]

NET WEIGHT 147 lb [67 kg]

ANGLE OF INCLINATION FROM THE VERTICAL 0° TO 60°

MIN. CREEP 29.50 [749.3] MIN. STRIKE 14.00 [355.6]

CREEP/L-G 1.48 [37.6] STD. IEEE C57.19.00/01

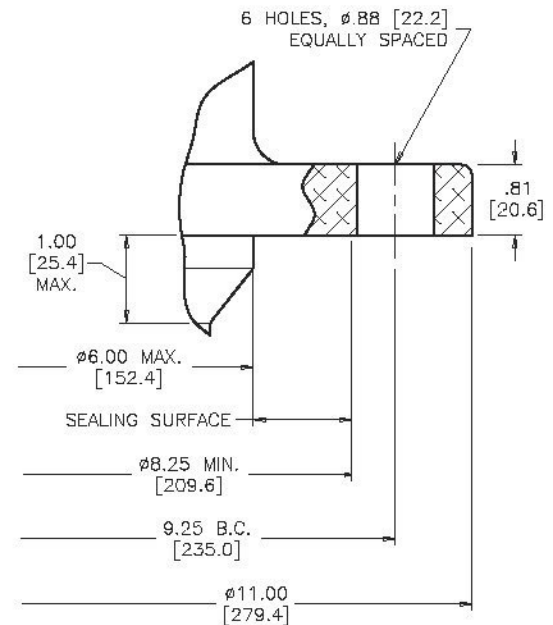
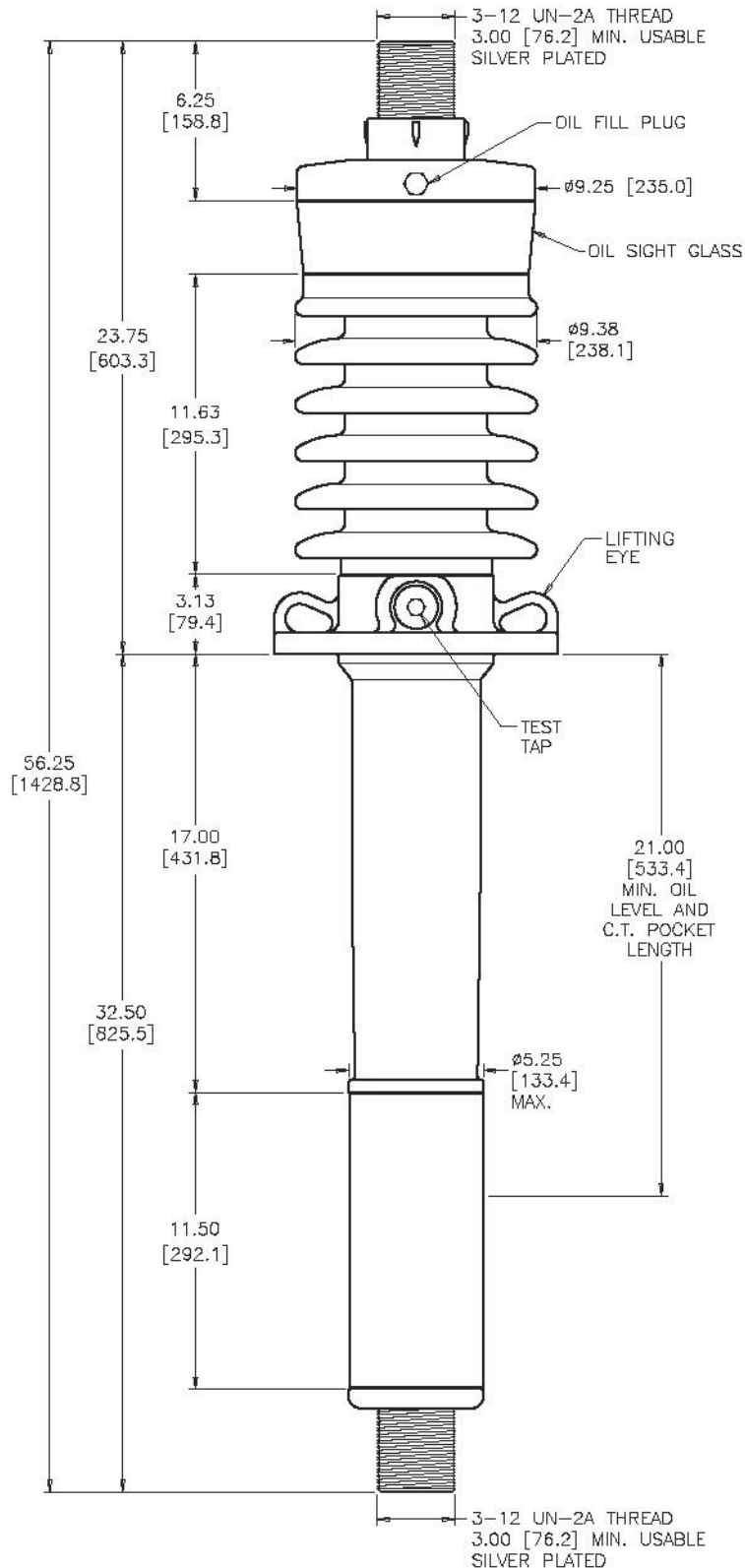
PCB CONTENT OF OIL IS LESS THAN 1 ppm

TITLE

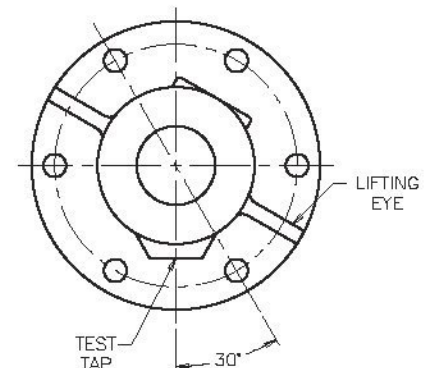
POC BUSHING

125°C INSULATION

VTC PART# 55734239



MOUNTING FLANGE DETAIL



**FLANGE ALIGNMENT DIAGRAM
TOP VIEW**

DIMENSIONS IN INCHES [MILLIMETERS]

REVISION CREEP/L-G WAS CREEP/U_y, REV. DESCRIPTION

SCALE 1/5

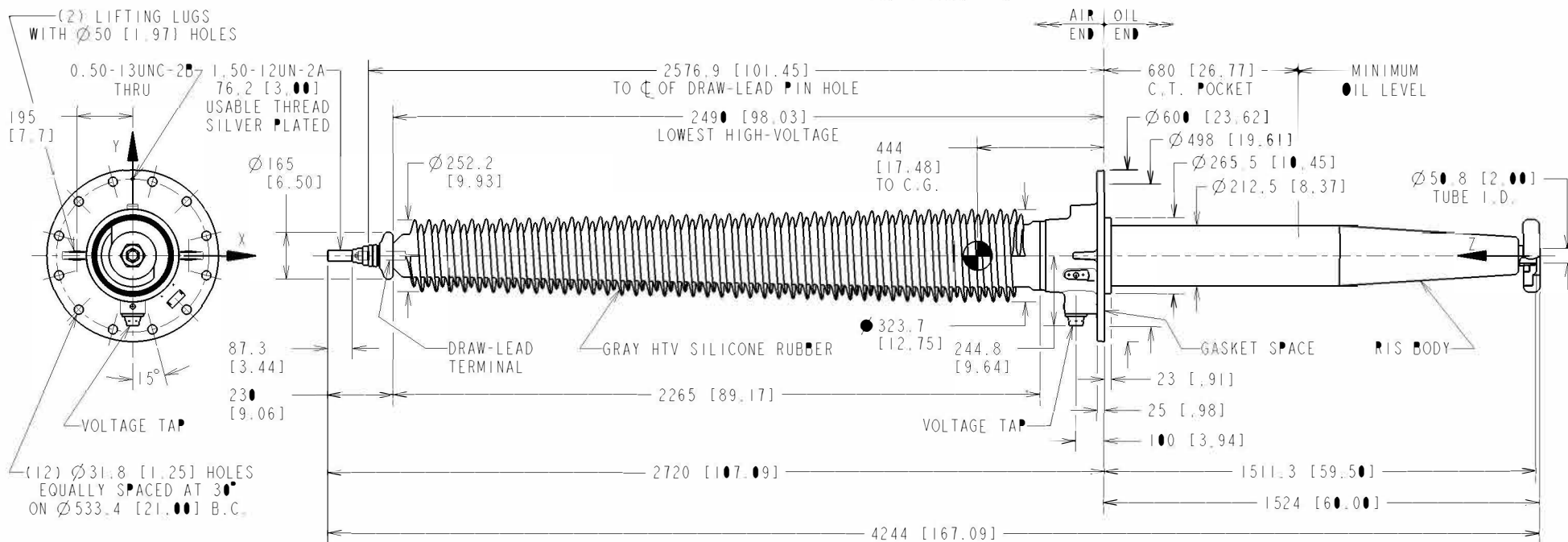
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SEISMIC OUTLINE DRAWING

QUALIFIED TO: IEEE Std 693-2005 LEVEL: HIGH P.L.
 REPORT NUMBER: PEER/STI 2017-19 TEST DATE: May 17, 2017
 QUALIFIED BY: TIME HISTORY SHAKE TABLE TEST
 A REPORT APPROVED BY: AMIR GILANI, PhD
 NAT. FREQUENCY & DAMPING: 7.8 ~ 8.1 Hz @ 4%
 MAX. RELATIVE DEFLECTION: 4.7 INCHES
 MAX. BOLT LOAD: UPWARD: 4.4 kips SHEAR: 0.5 kips

SPECIAL FEATURES: NONE

MAY BE CONVERTED TO INBOARD END CONNECTED
 BUSHING, STYLE 230N1200CA, BY CHANGING
 THE TOP ACCESSORIES AND BOTTOM SHIELD



* CURRENT RATING AND THERMAL CONSTANTS BASED ON USE OF A 1,000 MCM COPPER CABLE WITH 1.5mm PAPER INSULATION, AND AN OIL LEVEL INSIDE THE TUBE AT LEAST 1/3 OF THE AIR END HEIGHT.

INSULATION CLASS kV	BASIC IMPULSE LEVEL kV	MAX. LINE TO GROUND VOLTAGE kV	CONNECTION METHOD AND CONTINUOUS CURRENT A @ 60Hz*			THERMAL CONSTANTS*				CREEPAGE DISTANCE mm / [in]		MINIMUM ARCING DISTANCE mm / [in]	MAXIMUM ALTITUDE m / [ft]	MAX. CONTINUOUS CANTILEVER LOAD N / [lbf]		MOUNTING ANGLE FROM VERTICAL deg	APPROX. WEIGHT kg / [lbf]
			DRAW-LEAD	INBOARD END		K1	K2	n	τ	NOMINAL	MINIMUM			OUTBOARD	INBOARD		
230	1050	146	800	N/A	N/A	21.7	0.89	2.33	80	8790 [346.1]	8525 [335.6]	2265.0 [89.17]	2591 [8500]	3000 [675]	3000 [675]	0 TO 90	242.6 [535]

REVISEMENTS	NO. & LOC.	- ACTIVE - NEW ISSUE	ABB Inc. ALAMO, TN U.S.A.										ABB P/E FILE KEY: Plus Dry								
			TITLE: TYPE Plus Dry™, IEEE, DRY CONDENSER BUSHING																		
			RESP. DEPT.: BUSHINGS					REF. =													
			DFT: L. ELDER					8/3/2016					UNITS = mm/[in]								
			CHKD: W.A. SHINER					8/15/2016					SCALE = 1:1								
- ACTIVE - SEISMIC TABLE AND THERMAL CONSTANTS INFORMATION UPDATED W.A. SHINER 11-21-17 L. ELDER 11-27-17	B	APPD: R. NADASKAY												9/12/2016		LANG. = ENGLISH		REV: 0		SHEET 1 OF 1	